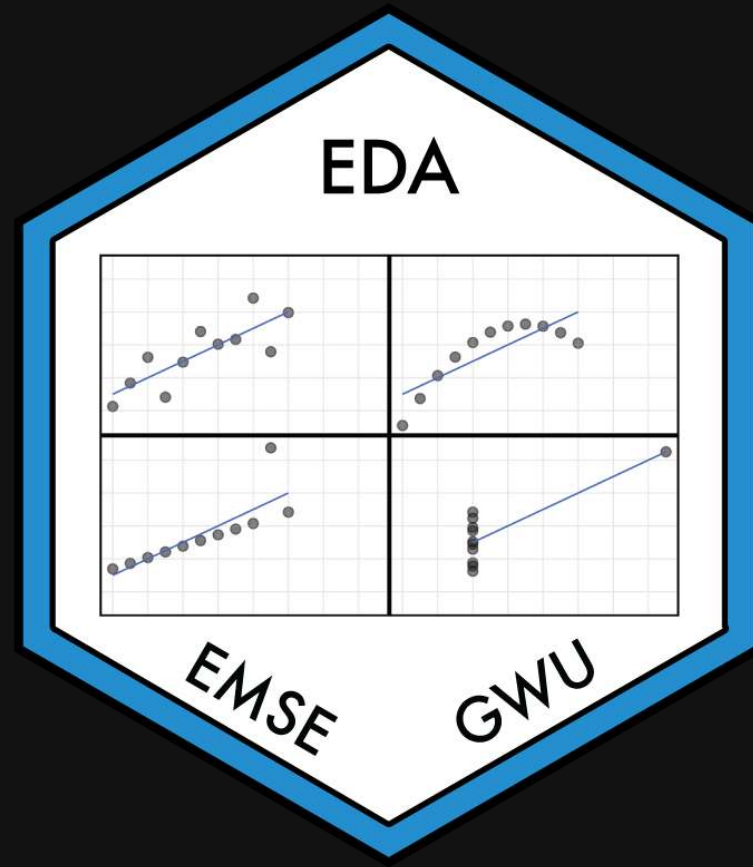




Exploring Data

🏛️ EMSE 4572/6572: Exploratory Data Analysis

👤 John Paul Helveston

📅 September 30, 2026

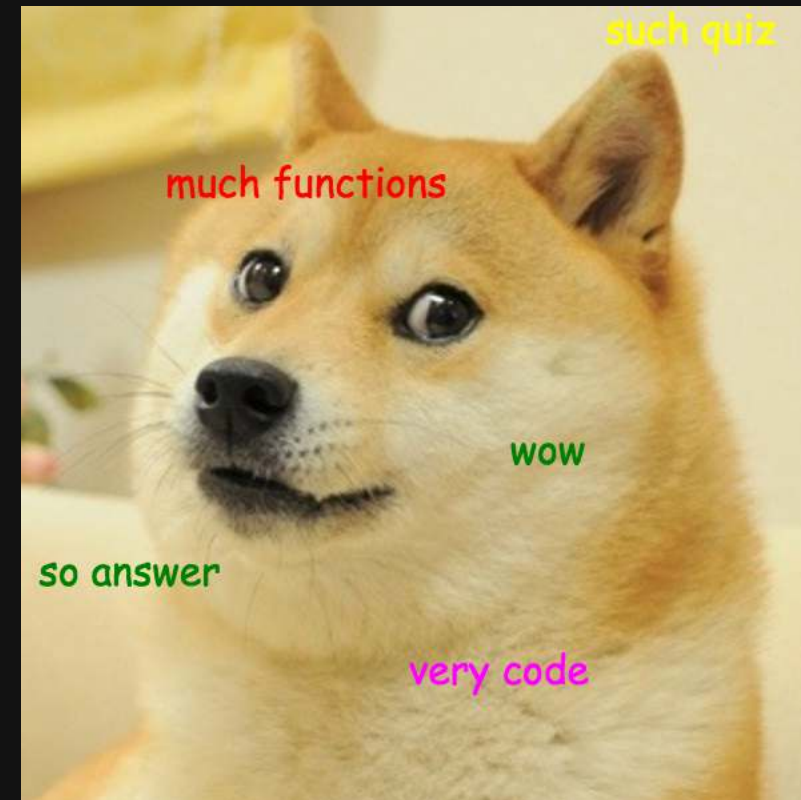
Quiz 2

10:00

Write your name on the quiz!

Rules:

- Work alone; no outside help of any kind is allowed.
- No calculators, no notes, no books, no computers, no phones.

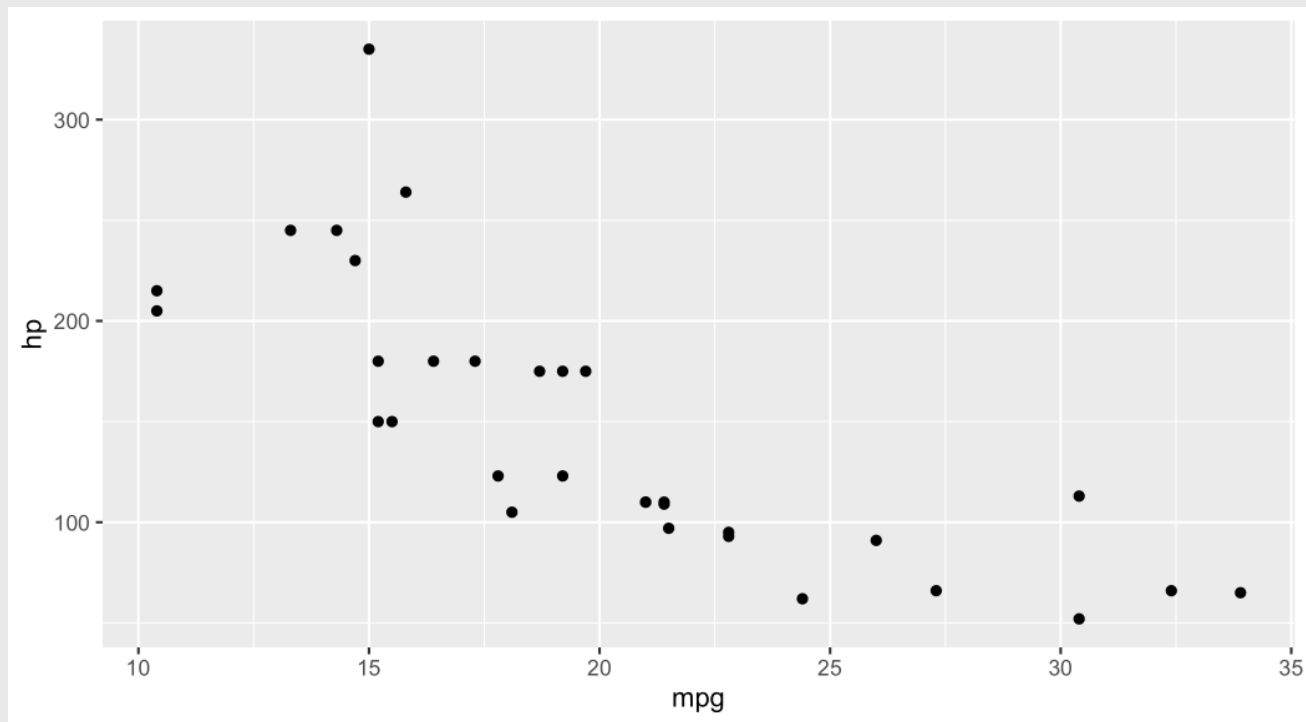


Tip of the week:

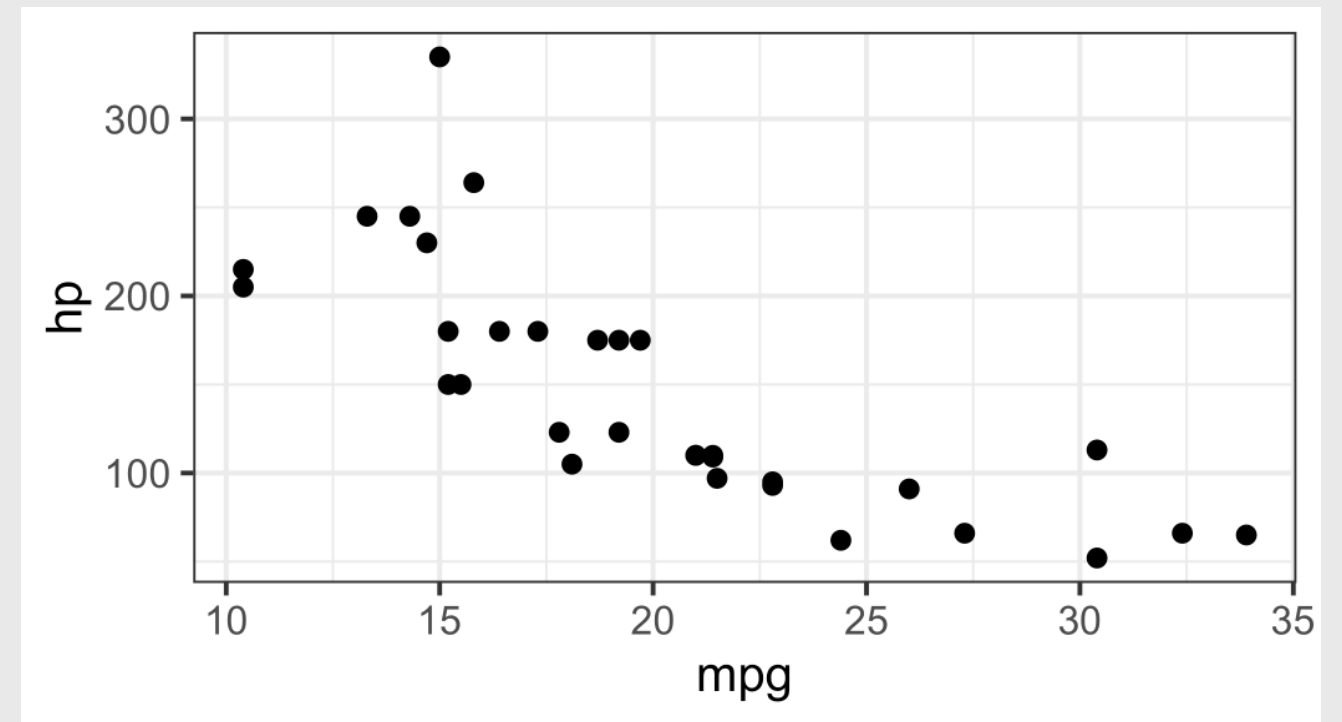
`theme_set()`

```
1 ggplot(mtcars) +  
2   geom_point(aes(x = mpg, y = hp))
```

Default theme



`theme_bw(base_size = 20)`



Week 6: *Exploring Data*

1. Exploring Data
2. Data Types
3. Centrality & Variability
4. Visualizing Centrality & Variability
5. Correlation
6. Visualizing Correlation
7. Visualizing Relationships

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Exploratory Analysis

Goal: **Form** hypotheses.

Improves quality of **questions**.

(what we do in THIS class)

Confirmatory Analysis

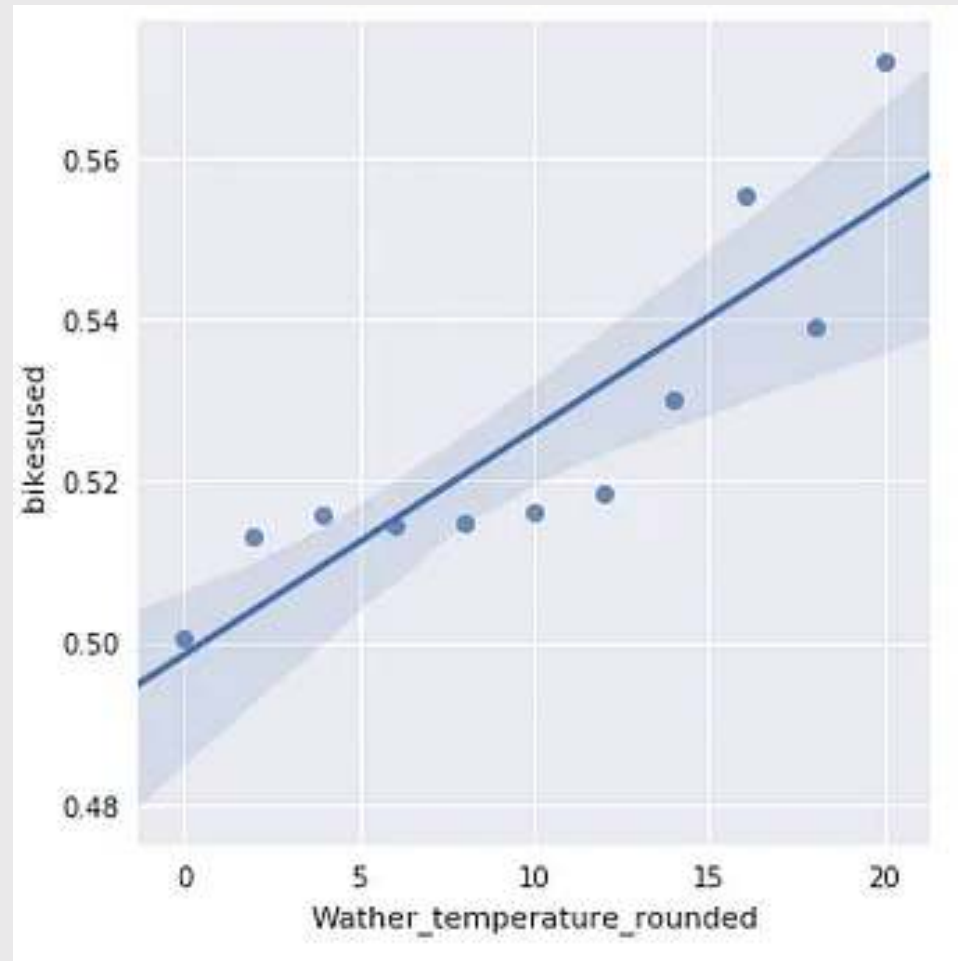
Goal: **Test** hypotheses.

Improves quality of **answers**.

(what you do in a stats class)

Exploratory Analysis

RQ: Do people bike more when the weather is nice?



Confirmatory Analysis

Let's build a model to predict bike usage based on weather.

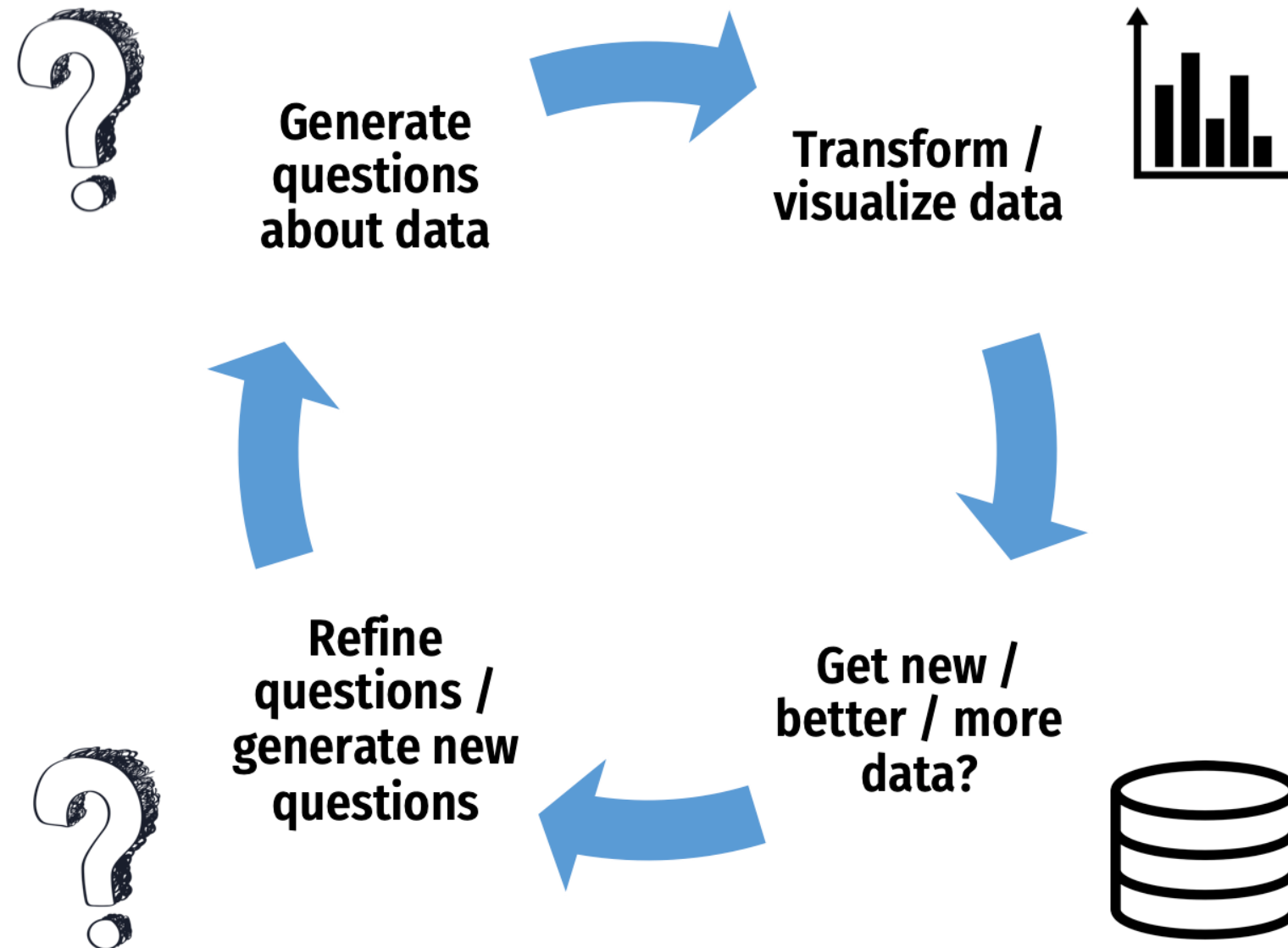
Don't be Icarus



“An *approximate* answer to the *right* question is better than an *exact* answer to the *wrong* question.”

— [John Tukey](#).

EDA is an iterative process to help you
understand your data and ask better questions



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24,901

Earth's circumference at the equator:
24,901

Earth's circumference at the equator:
24,901 miles

Types of Data

Categorical

Subdivide things into *groups*

- What type?
- Which category?

Numerical

Measure things with numbers

- How many?
- How much?

Categorical (discrete) variables

Nominal

- Order doesn't matter
- Differ in "name" (nominal) only

e.g. `country` in TB case data:

```
#> # A tibble: 6 × 4
#>   country      year  cases population
#>   <chr>      <dbl> <dbl>      <dbl>
#> 1 Afghanistan 1999     745    19987071
#> 2 Afghanistan 2000    2666    20595360
#> 3 Brazil      1999   37737    172006362
#> 4 Brazil      2000   80488    174504898
#> 5 China       1999  212258   1272915272
#> 6 China       2000  213766   1280428583
```

Ordinal

- Order matters
- Distance between units not equal

e.g.: `Placement` 2017 Boston marathon:

```
#> # A tibble: 6 × 3
#>   Placement `Official Time` Name
#>   <dbl> <time>      <chr>
#> 1      1 02:09:37 Kirui, Geoffrey
#> 2      2 02:09:58 Rupp, Galen
#> 3      3 02:10:28 Osako, Suguru
#> 4      4 02:12:08 Biwott, Shadrack
#> 5      5 02:12:35 Chebet, Wilson
#> 6      6 02:12:45 Abdirahman, Abdi
```

Numerical data

Interval

- Numerical scale with arbitrary starting point
- No "0" point
- Can't say "x" is double "y"

e.g.: `temp` in Beaver data

```
#>   day time  temp activ
#> 1 346  840 36.33     0
#> 2 346  850 36.34     0
#> 3 346  900 36.35     0
#> 4 346  910 36.42     0
#> 5 346  920 36.55     0
#> 6 346  930 36.69     0
```

Ratio

- Has a "0" point
- Can be described as percentages
- Can say "x" is double "y"

e.g.: `height` & `speed` in wildlife impacts

```
#> # A tibble: 6 × 3
#>   incident_date      height speed
#>   <dtm>          <dbl> <dbl>
#> 1 2018-12-31 00:00:00    700    200
#> 2 2018-12-27 00:00:00    600    145
#> 3 2018-12-23 00:00:00     0    130
#> 4 2018-12-22 00:00:00    500    160
#> 5 2018-12-21 00:00:00    100    150
#> 6 2018-12-18 00:00:00   4500    250
```

Key Questions

Categorical

Does the order matter?

Yes: **Ordinal**

No: **Nominal**

Numerical

Is there a “baseline”?

Yes: **Ratio**

No: **Interval**

Be careful of how variables are encoded!

When numbers are categories

- "Dummy coding": e.g., `passedTest = 1` or `0`)
- "North", "South", "East", "West" = `1, 2, 3, 4`

When ratio data are discrete (i.e. counts)

- Number of eggs in a carton, heart beats per minute, etc.
- Continuous variables measured discretely (e.g. age)

Time

- As *ordinal* categories: "Jan.", "Feb.", "Mar.", etc.
- As *interval* scale: "Jan. 1", "Jan. 2", "Jan. 3", etc.
- As *ratio* scale: "30 sec", "60 sec", "70 sec", etc.

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Summary Measures:

Single variables: **Centrality** & **Variability**

Two variables: **Correlation**

Centrality (a.k.a. The “Average” Value)

A single number representing the *middle* of a set of numbers

Mean: $\frac{\text{Sum of values}}{\text{\# of values}}$

Median: “Middle” value (50% of data above & below)

Mean isn't always the “best” choice

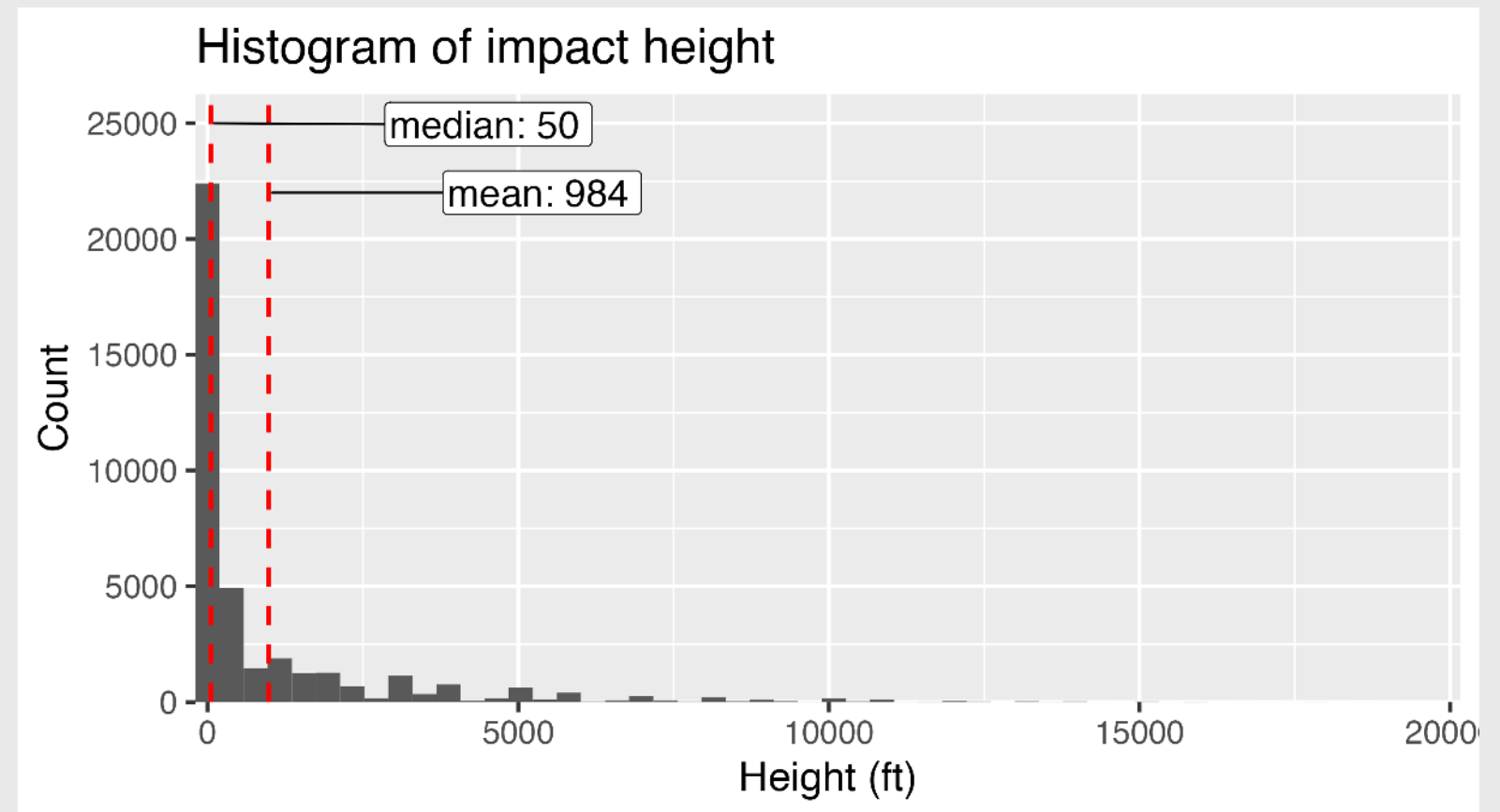
```
1 wildlife_impacts %>%
2   filter(!is.na(height)) %>%
3   summarise(
4     mean = mean(height),
5     median = median(height)
6   )
```

```
#> # A tibble: 1 × 2
#>   mean median
#>   <dbl>  <dbl>
#> 1  984.    50
```

Percent of data below mean:

```
#> [1] "73.9%"
```

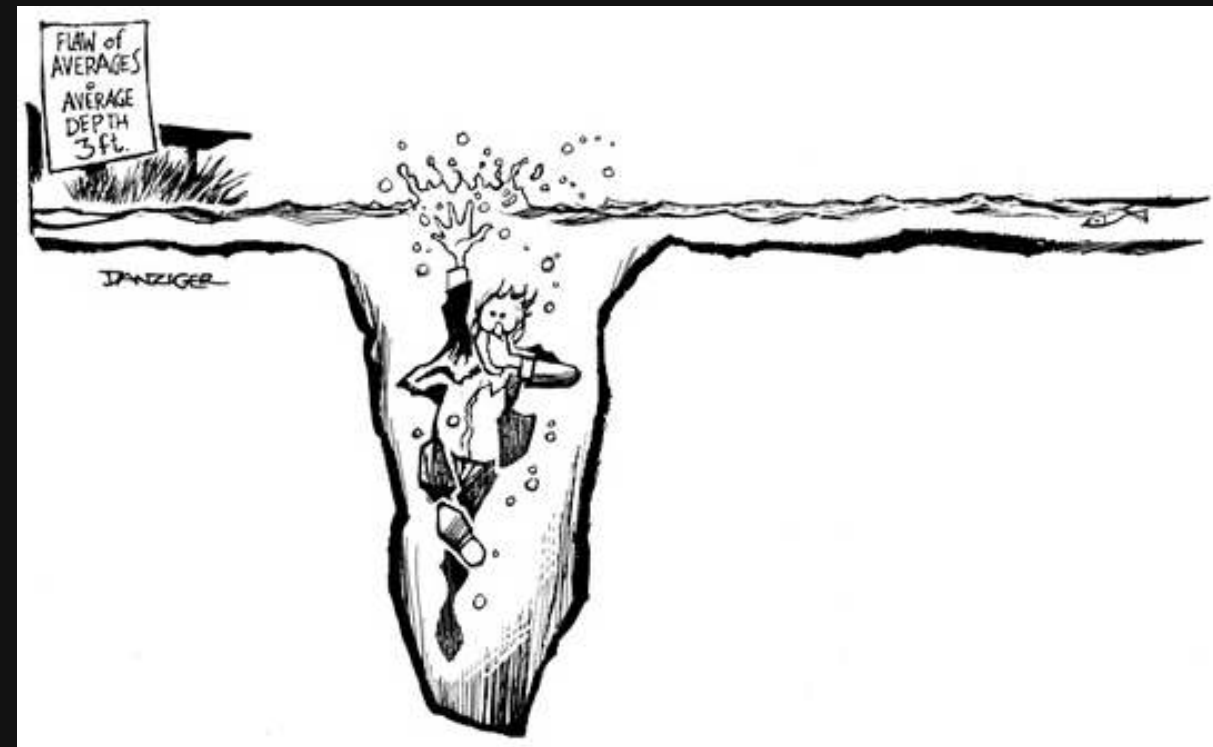
On average, at what height do planes hit birds?



Beware the “flaw of averages”

What happened to the statistician
that crossed a river with an average
depth of 3 feet?

...he drowned



Variability (“Spread”)

Standard deviation: distribution of values relative to the mean

$$s = \sqrt{\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N-1}}$$

Interquartile range (IQR): $Q_3 - Q_1$ (middle 50% of data)

Range: max - min

Example: Days to ship

Complaints are coming in about orders shipped from warehouse B, so you collect some data:

```
1 daysToShip
```

```
#>   order warehouseA warehouseB
#> 1      1           3           1
#> 2      2           3           1
#> 3      3           3           1
#> 4      4           4           3
#> 5      5           4           3
#> 6      6           4           4
#> 7      7           5           5
#> 8      8           5           5
#> 9      9           5           5
#> 10    10           5           6
#> 11    11           5           7
#> 12    12           5          10
```

Here, **averages** are misleading:

```
1 daysToShip %>%
2   gather(warehouse, days, warehouseA:warehouseB) %>%
3   group_by(warehouse) %>%
4   summarise(
5     mean = mean(days),
6     median = median(days)
7   )
```

```
#> # A tibble: 2 × 3
#>   warehouse    mean median
#>   <chr>      <dbl>  <dbl>
#> 1 warehouseA  4.25    4.5
#> 2 warehouseB  4.25    4.5
```

Example: Days to ship

Complaints are coming in about orders shipped from warehouse B, so you collect some data:

```
1 daysToShip
```

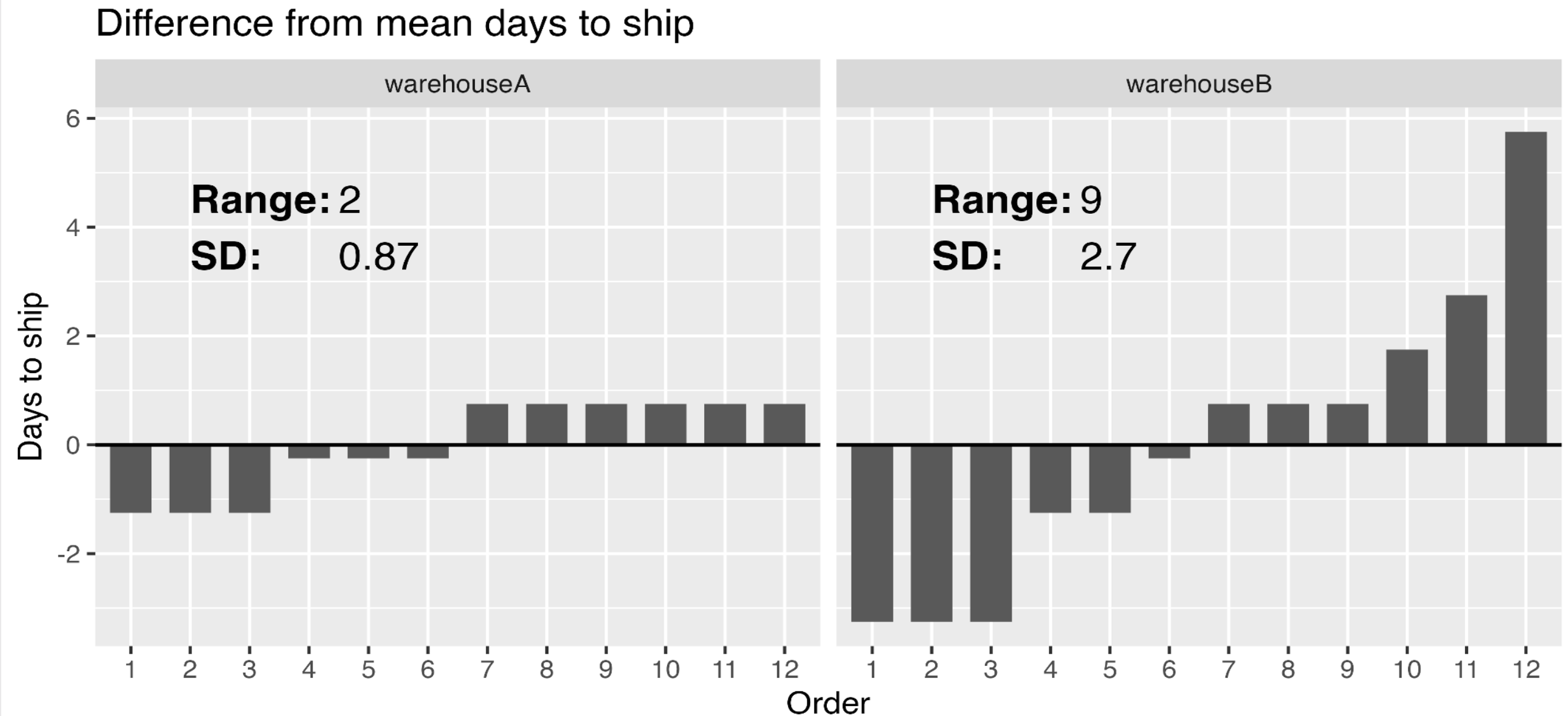
```
#>   order warehouseA warehouseB
#> 1      1           3           1
#> 2      2           3           1
#> 3      3           3           1
#> 4      4           4           3
#> 5      5           4           3
#> 6      6           4           4
#> 7      7           5           5
#> 8      8           5           5
#> 9      9           5           5
#> 10    10           5           6
#> 11    11           5           7
#> 12    12           5          10
```

Variability reveals difference in days to ship:

```
1 daysToShip %>%
2   gather(warehouse, days, warehouseA:warehouseB) %>%
3   group_by(warehouse) %>%
4   summarise(
5     mean = mean(days),
6     median = median(days),
7     range = max(days) - min(days),
8     sd = sd(days)
9   )
```

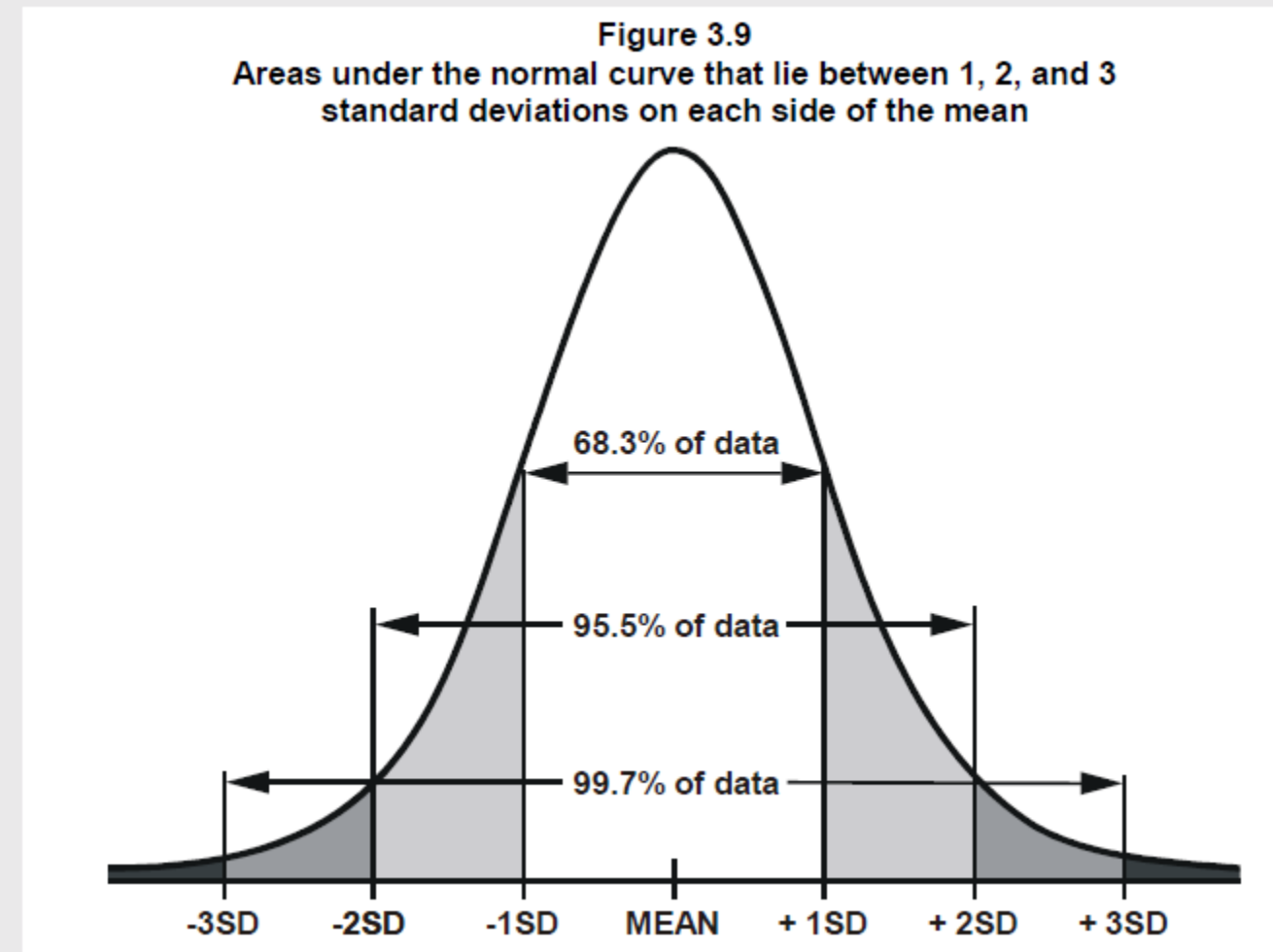
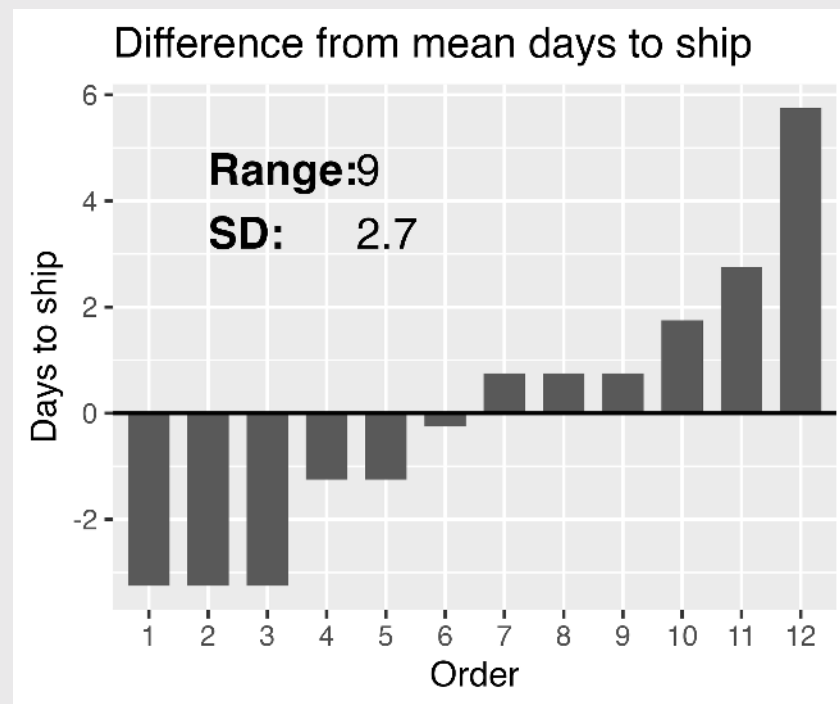
```
#> # A tibble: 2 × 5
#>   warehouse    mean median  range    sd
#>   <chr>      <dbl>  <dbl> <dbl> <dbl>
#> 1 warehouseA  4.25    4.5    2 0.866
#> 2 warehouseB  4.25    4.5    9 2.70
```

Example: Days to ship



Interpreting the standard deviation

$$S = \sqrt{\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N-1}}$$



Outliers



Mean & Standard Deviation are sensitive to outliers

Outliers: $Q_1 - 1.5IQR$ or $Q_3 + 1.5IQR$

Extreme values: $Q_1 - 3IQR$ or $Q_3 + 3IQR$

Some data:

```
1 data1 <- c(3, 3, 4, 5, 5, 6, 6, 7, 8, 9)
```

- Mean: 5.6
- Standard Deviation: 2.01
- Median: 5.5
- IQR: 2.5

Add just one extreme value:

```
1 data2 <- data1  
2 data2[10] <- 20
```

- Mean: 6.7
- Standard Deviation: 4.95
- Median: 5.5
- IQR: 2.5

Robust statistics for continuous data (less sensitive to outliers)

Centrality: Use *median* rather than *mean*

Variability: Use *IQR* rather than *standard deviation*

Practice with summary measurements

10:00

1) Read in the following data sets:

- `milk_production.csv`
- `lotr_words.csv`

2) For each variable in each data set, if possible, summarize its

1. **Centrality**
2. **Variability**

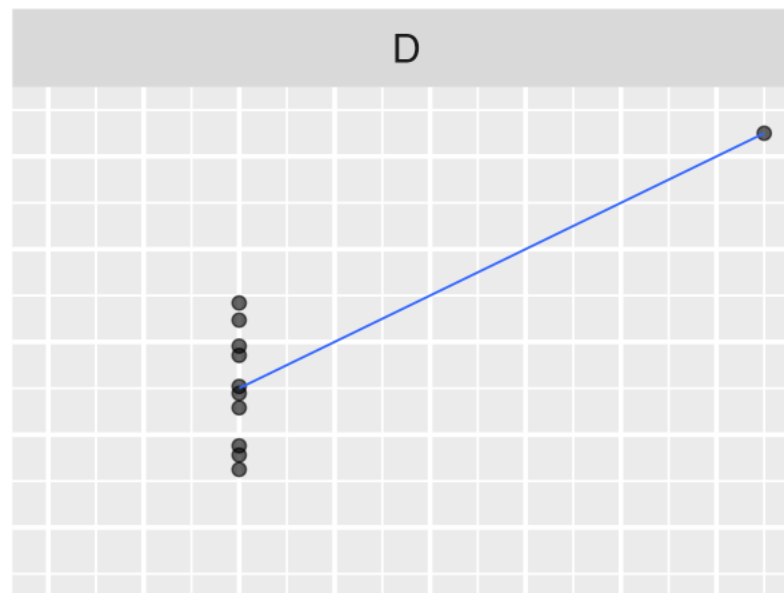
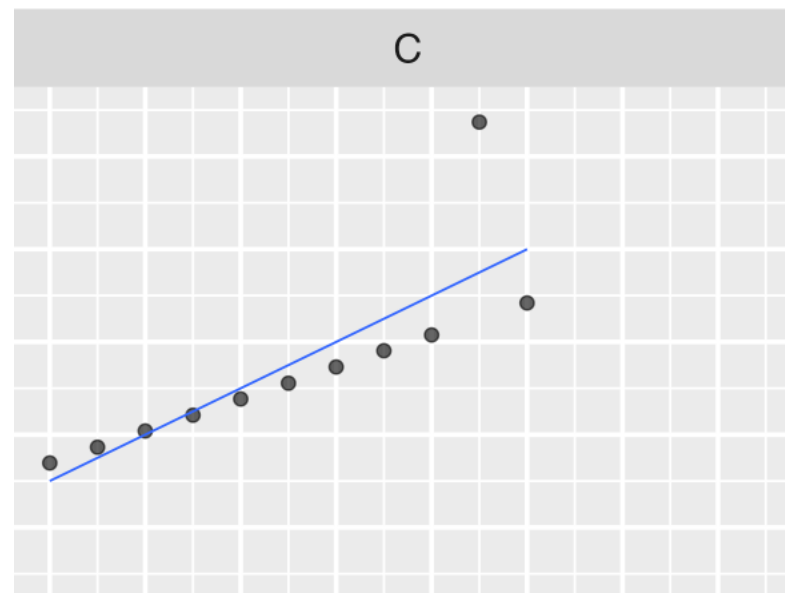
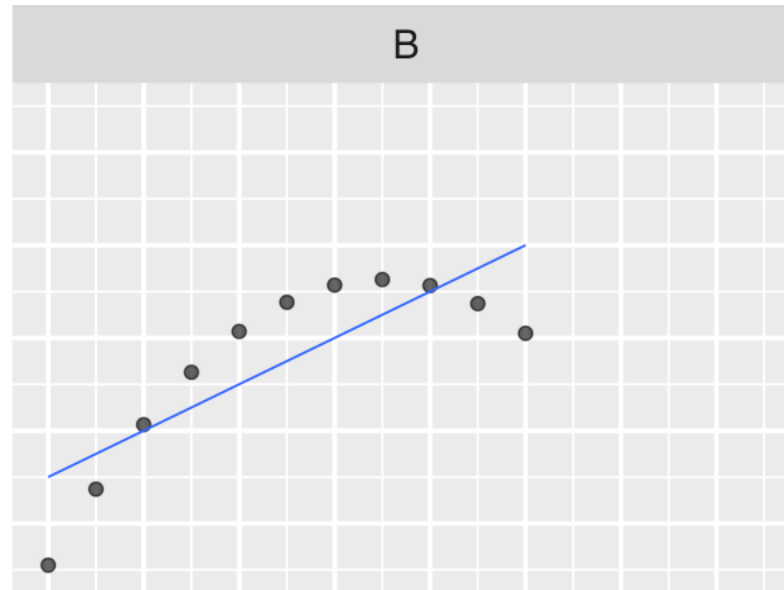
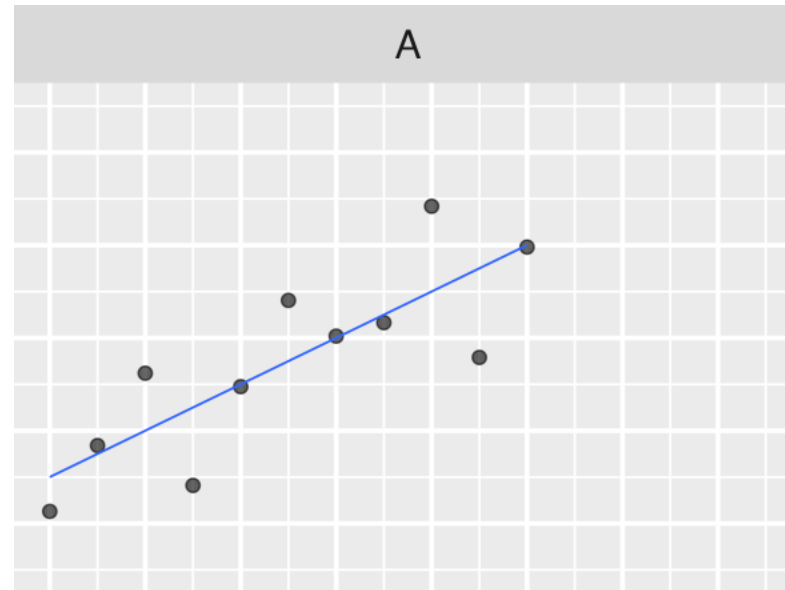
Week 6: *Exploring Data*

1. Exploring Data
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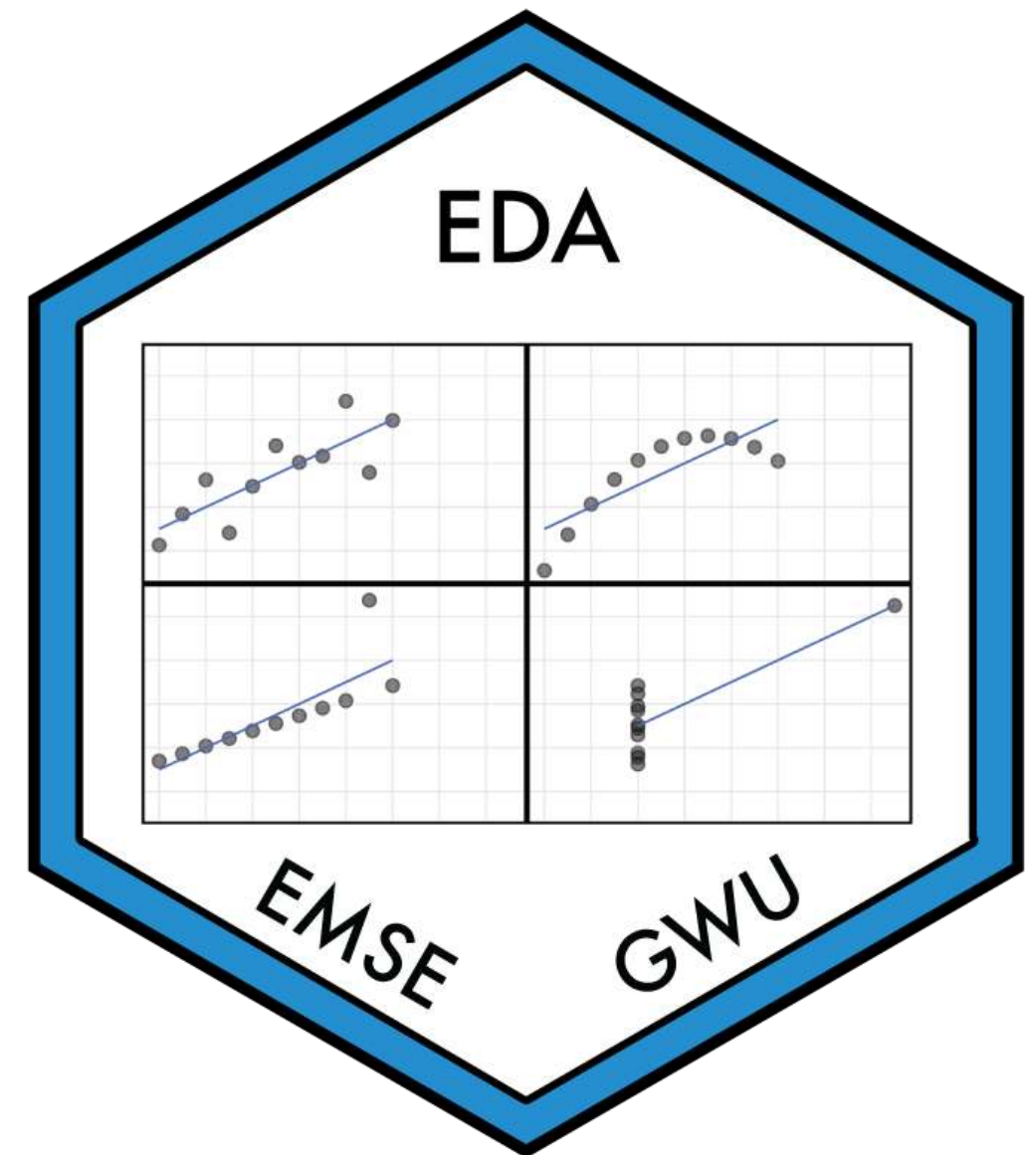
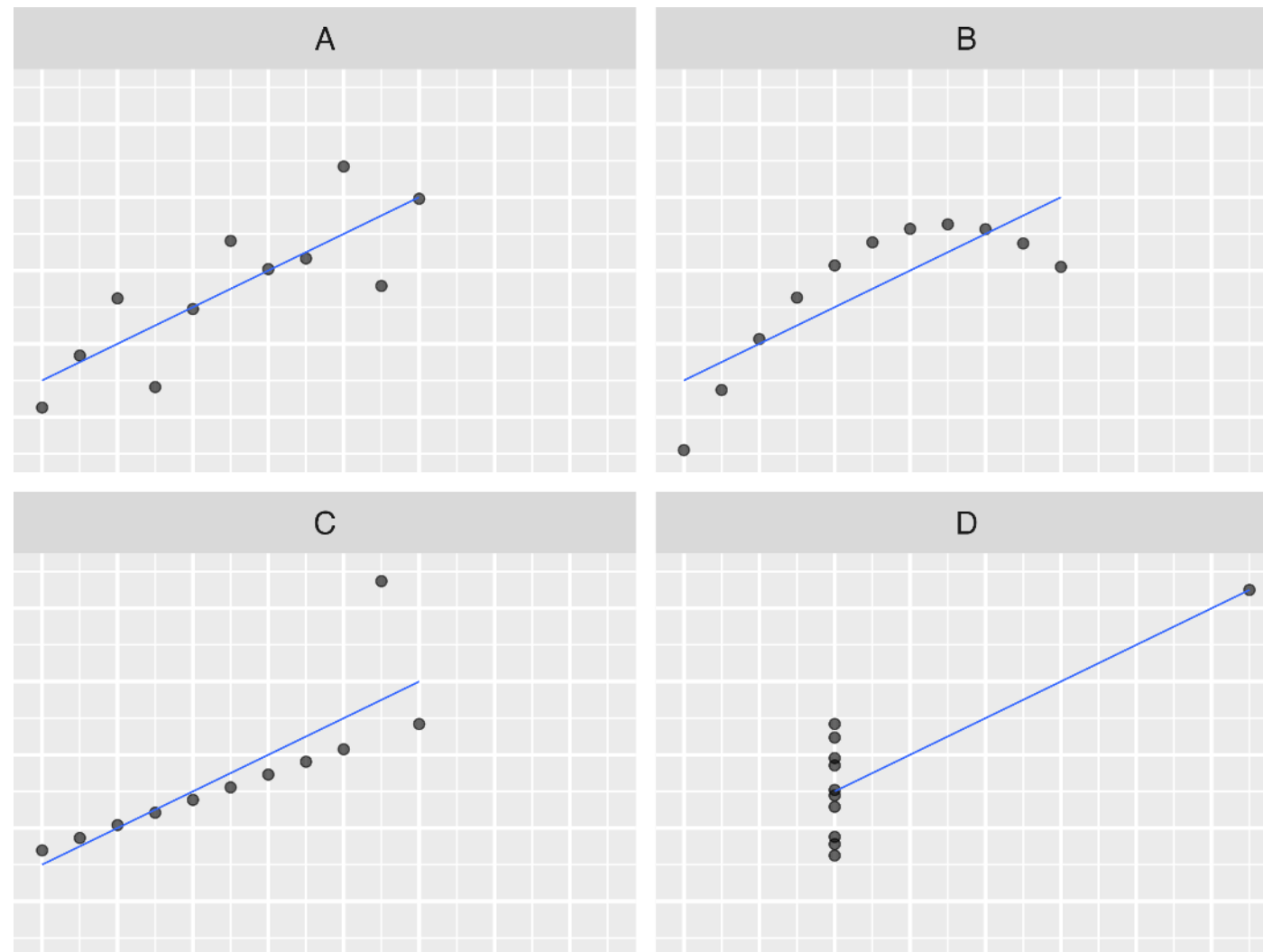
“Visualizing data helps us think”

	A		B		C		D	
	x	y	x	y	x	y	x	y
	10	8.04	10	9.14	10	7.46	8	6.58
	8	6.95	8	8.14	8	6.77	8	5.76
	13	7.58	13	8.74	13	12.74	8	7.71
	9	8.81	9	8.77	9	7.11	8	8.84
	11	8.33	11	9.26	11	7.81	8	8.47
	14	9.96	14	8.1	14	8.84	8	7.04
	6	7.24	6	6.13	6	6.08	8	5.25
	4	4.26	4	3.1	4	5.39	19	12.5
	12	10.84	12	9.13	12	8.15	8	5.56
	7	4.82	7	7.26	7	6.42	8	7.91
	5	5.68	5	4.74	5	5.73	8	6.89
Sum:	99	82.51	99	82.51	99	82.5	99	82.51
Mean:	9	7.5	9	7.5	9	7.5	9	7.5
St. Dev:	3.3	2	3.3	2	3.3	2	3.3	2

Anscombe's Quartet



Anscombe's Quartet



The data *type* determines
how to summarize it

Nominal (Categorical)

Measures:

- Frequency counts / Proportions

Charts:

- Bars

Ordinal (Categorical)

Measures:

- Frequency counts / Proportions
- **Centrality**: Median, Mode
- **Variability**: IQR

Charts:

- Bars

Numerical (Continuous)

Measures:

- **Centrality**: Mean, median
- **Variability**: Range, standard deviation, IQR

Charts:

- Histogram
- Boxplot

Summarizing **Nominal** data

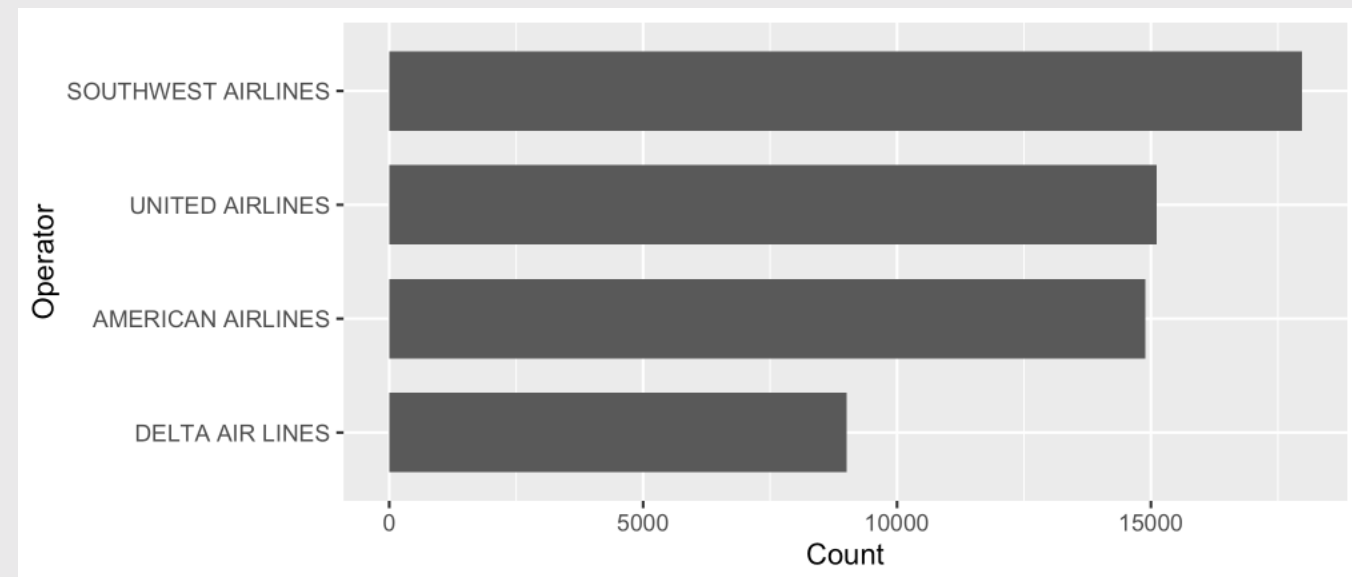
Summarize with counts / percentages

```
1 wildlife_impacts %>%
2   count(operator, sort = TRUE) %>%
3   mutate(p = n / sum(n))
```

```
#> # A tibble: 4 × 3
#>   operator          n      p
#>   <chr>        <int> <dbl>
#> 1 SOUTHWEST AIRLINES 17970 0.315
#> 2 UNITED AIRLINES   15116 0.265
#> 3 AMERICAN AIRLINES 14887 0.261
#> 4 DELTA AIR LINES    9005 0.158
```

Visualize with (usually sorted) bars

```
1 wildlife_impacts %>%
2   count(operator, sort = TRUE) %>%
3   ggplot() +
4   geom_col(
5     aes(x = n, y = reorder(operator, n)),
6     width = 0.7
7   ) +
8   labs(x = "Count", y = "Operator")
```



Summarizing Ordinal data

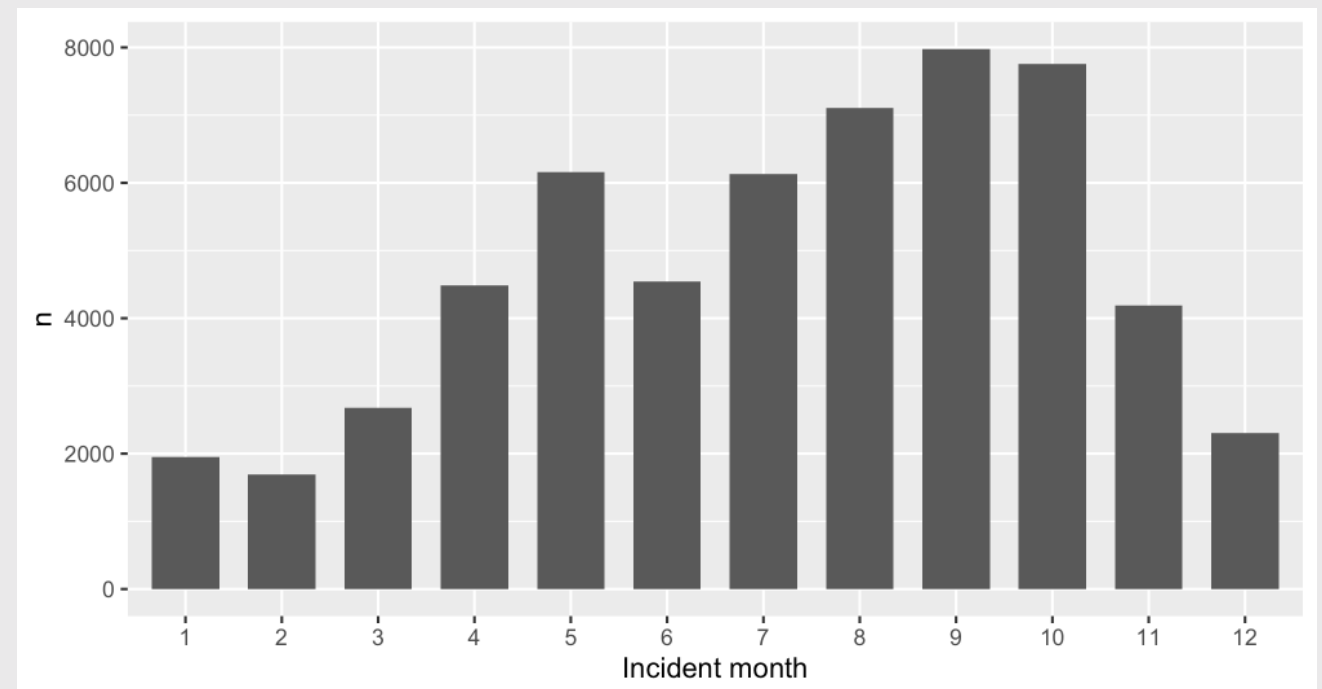
Summarize: Counts / percentages

```
1 wildlife_impacts %>%
2   count(incident_month, sort = TRUE) %>%
3   mutate(p = n / sum(n))
```

```
#> # A tibble: 12 × 3
#>   incident_month     n     p
#>   <dbl> <int> <dbl>
#> 1         9  7980 0.140
#> 2        10  7754 0.136
#> 3         8  7104 0.125
#> 4         5  6161 0.108
#> 5         7  6133 0.108
#> 6         6  4541 0.0797
#> 7         4  4490 0.0788
#> 8        11  4191 0.0736
#> 9         3  2678 0.0470
#> 10        12  2303 0.0404
#> 11         1  1951 0.0342
#> 12         2  1692 0.0297
```

Visualize: Bars

```
1 wildlife_impacts %>%
2   count(incident_month, sort = TRUE) %>%
3   ggplot() +
4   geom_col(
5     aes(x = as.factor(incident_month), y = n),
6     width = 0.7
7   ) +
8   labs(x = "Incident month")
```



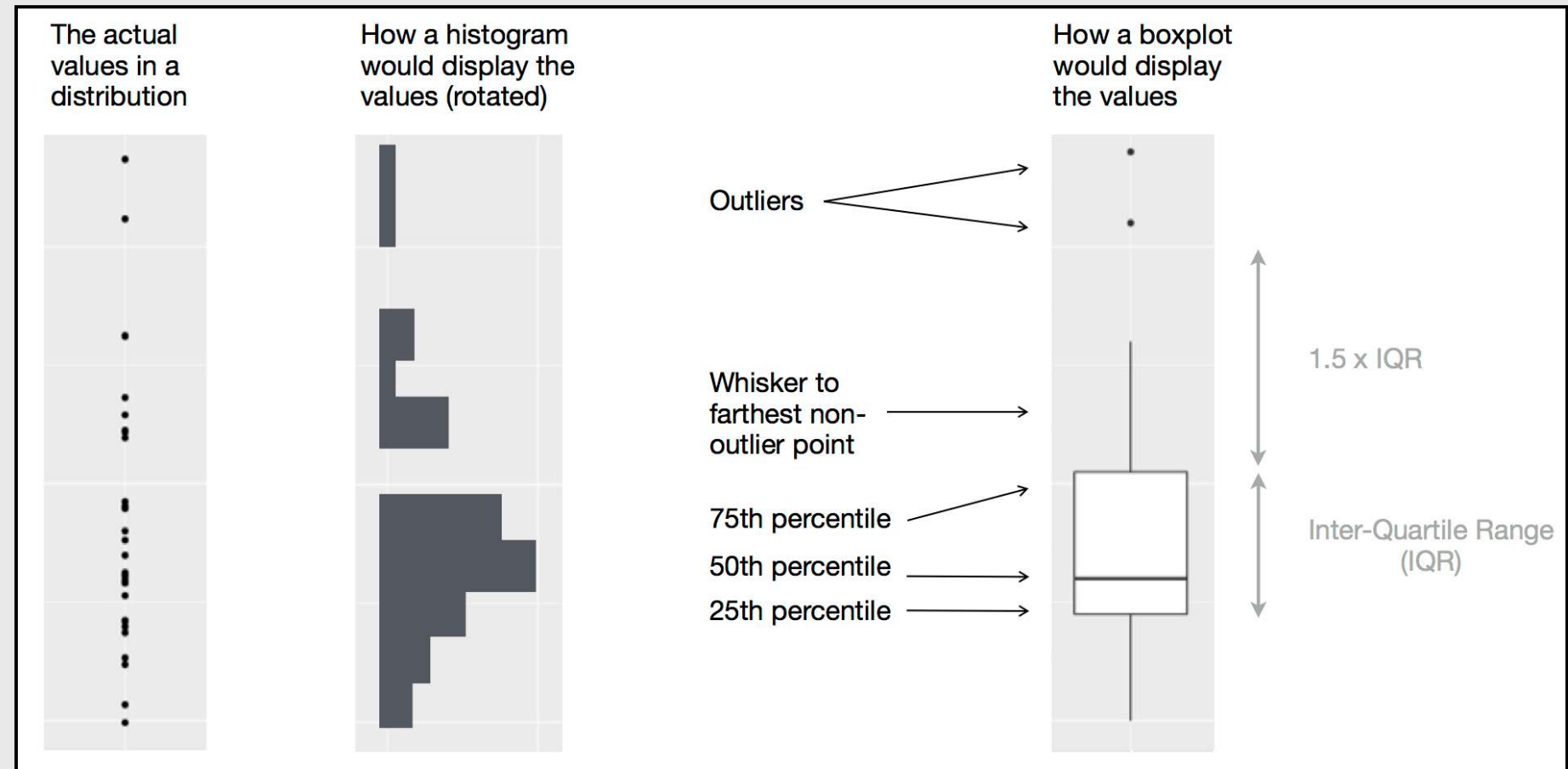
Summarizing **continuous** variables

Histograms:

- Skewness
- Number of modes

Boxplots:

- Outliers
- Comparing variables



Histogram: Identify Skewness & # of Modes

Summarise:

Mean, median, sd, range, & IQR:

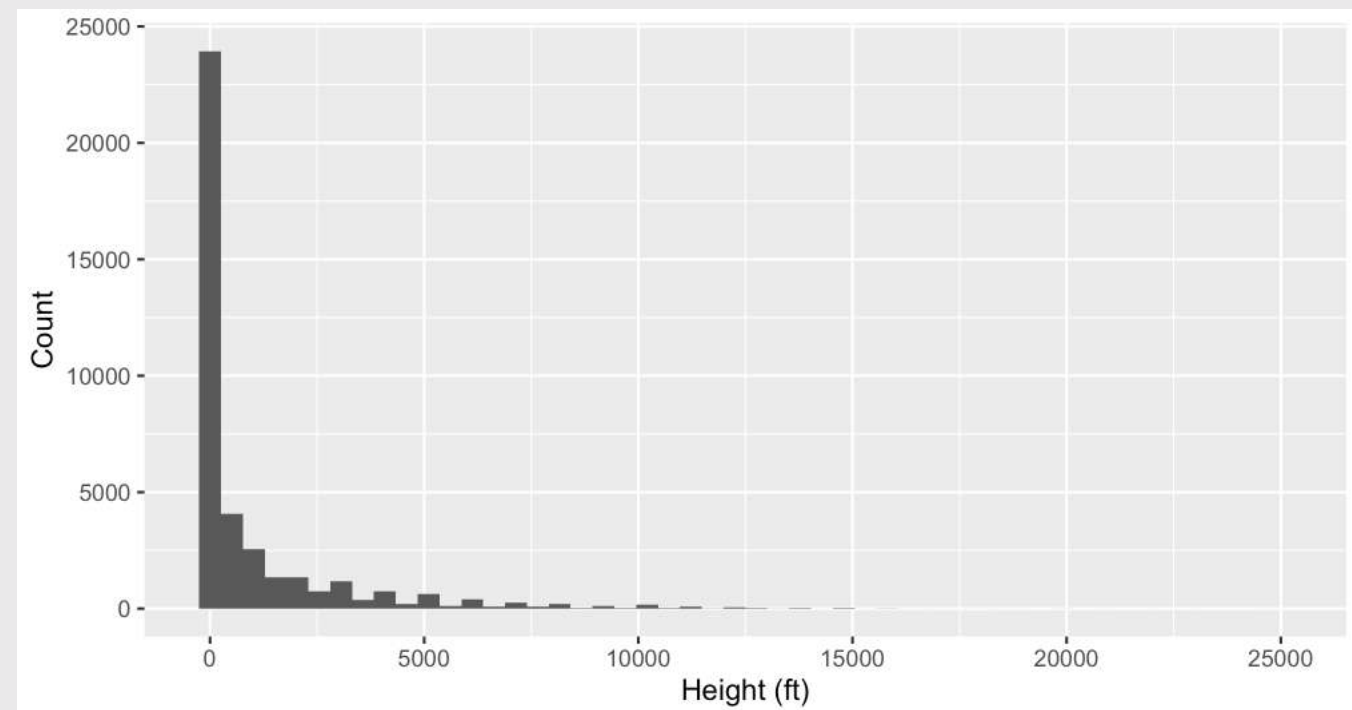
```
1 summary(wildlife_impacts$height)
```

```
#>      Min. 1st Qu.  Median    Mean 3rd Qu.
#>      0.0      0.0     50.0   983.8    100
```

Visualize:

Histogram (identify skewness & modes)

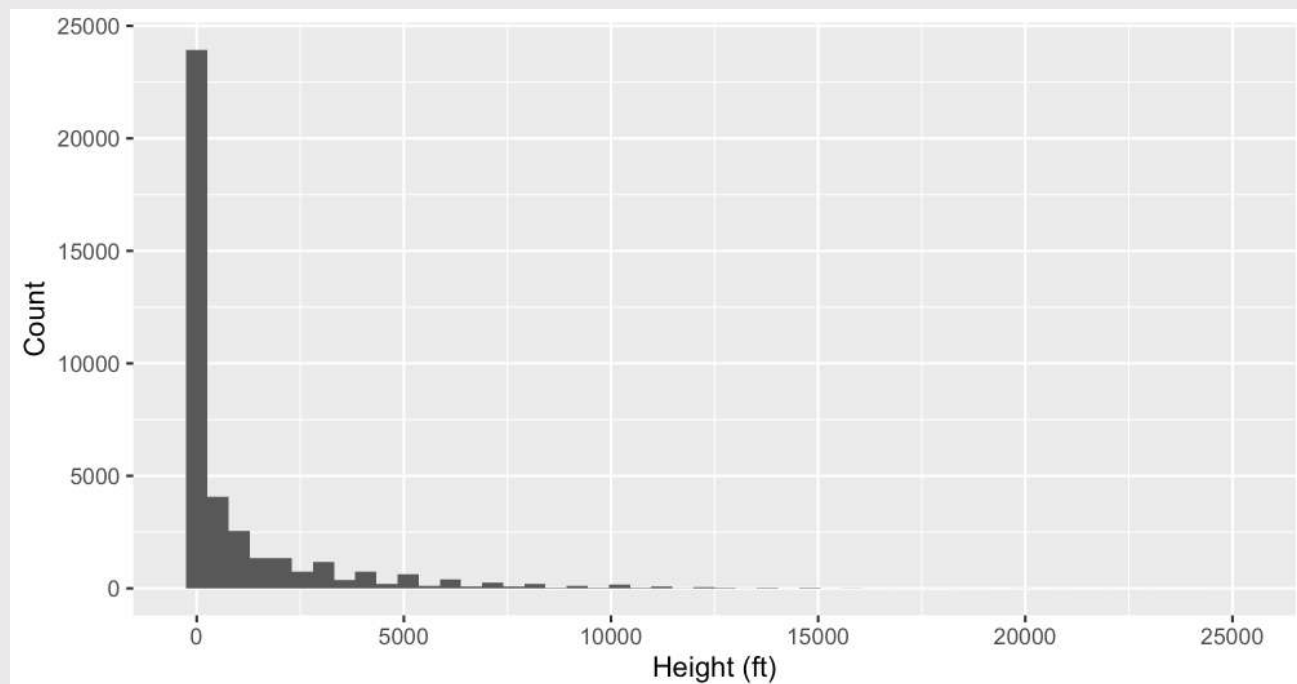
```
1 ggplot(wildlife_impacts) +
2   geom_histogram(
3     aes(x = height),
4     bins = 50
5   ) +
6   labs(x = 'Height (ft)', y = 'Count')
```



Histogram: Identify Skewness & # of Modes

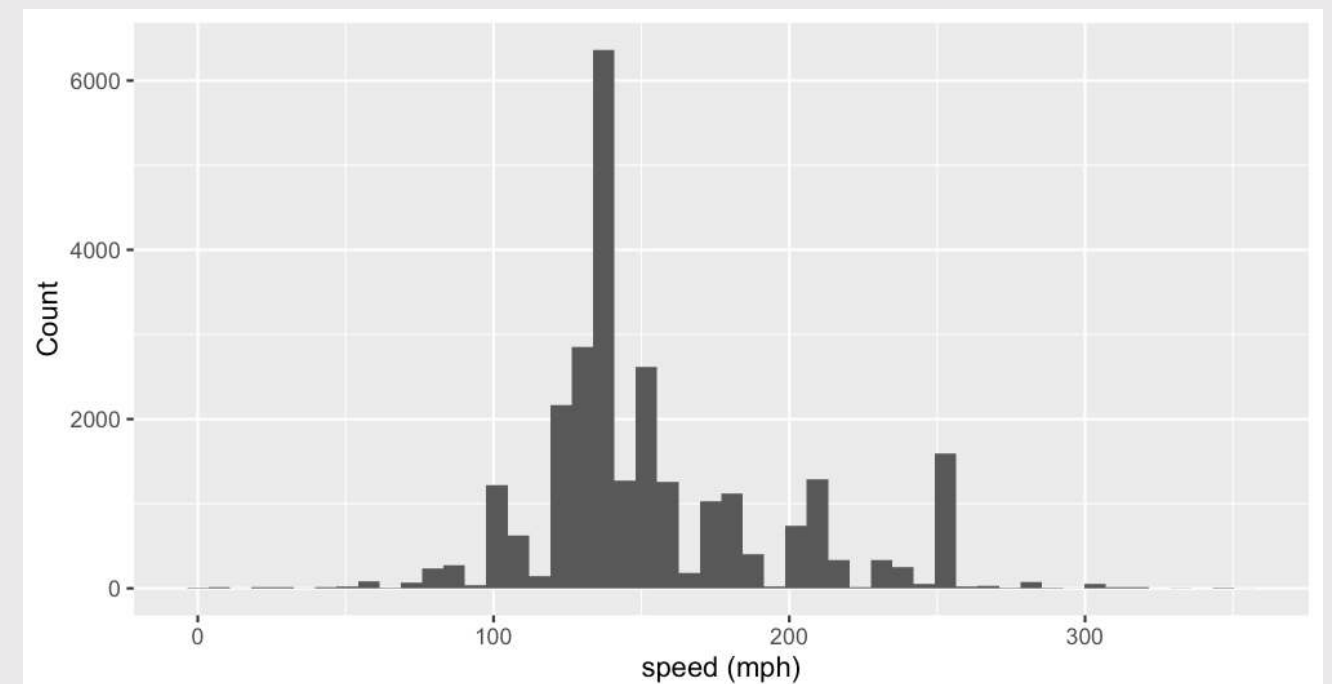
Height

```
1 ggplot(wildlife_impacts) +  
2   geom_histogram(  
3     aes(x = height),  
4     bins = 50  
5   ) +  
6   labs(x = 'Height (ft)', y = 'Count')
```



Speed

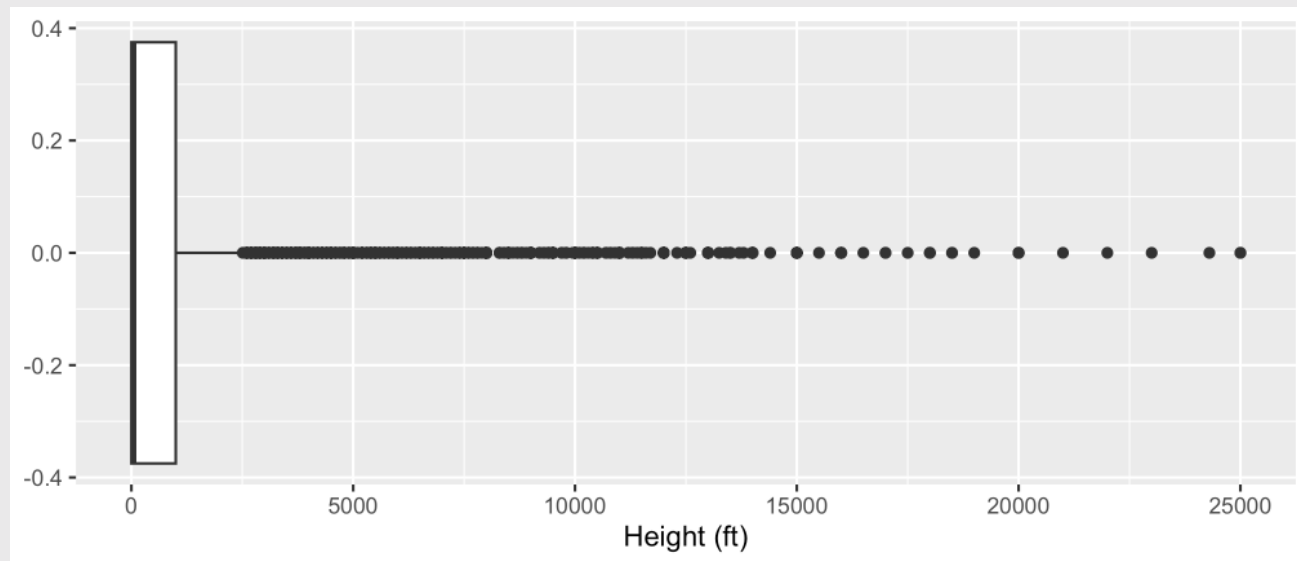
```
1 ggplot(wildlife_impacts) +  
2   geom_histogram(  
3     aes(x = speed),  
4     bins = 50  
5   ) +  
6   labs(x = 'speed (mph)', y = 'Count')
```



Boxplot: Identify outliers

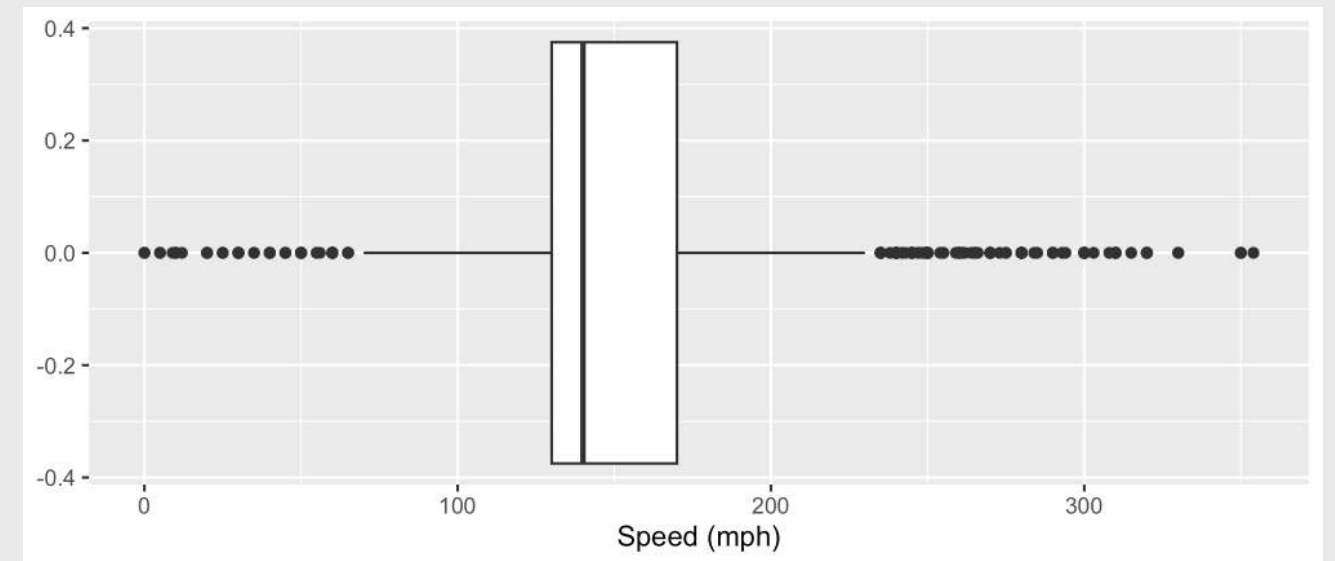
Height

```
1 ggplot(wildlife_impacts) +  
2   geom_boxplot(aes(x = height)) +  
3   labs(x = 'Height (ft)', y = NULL)
```



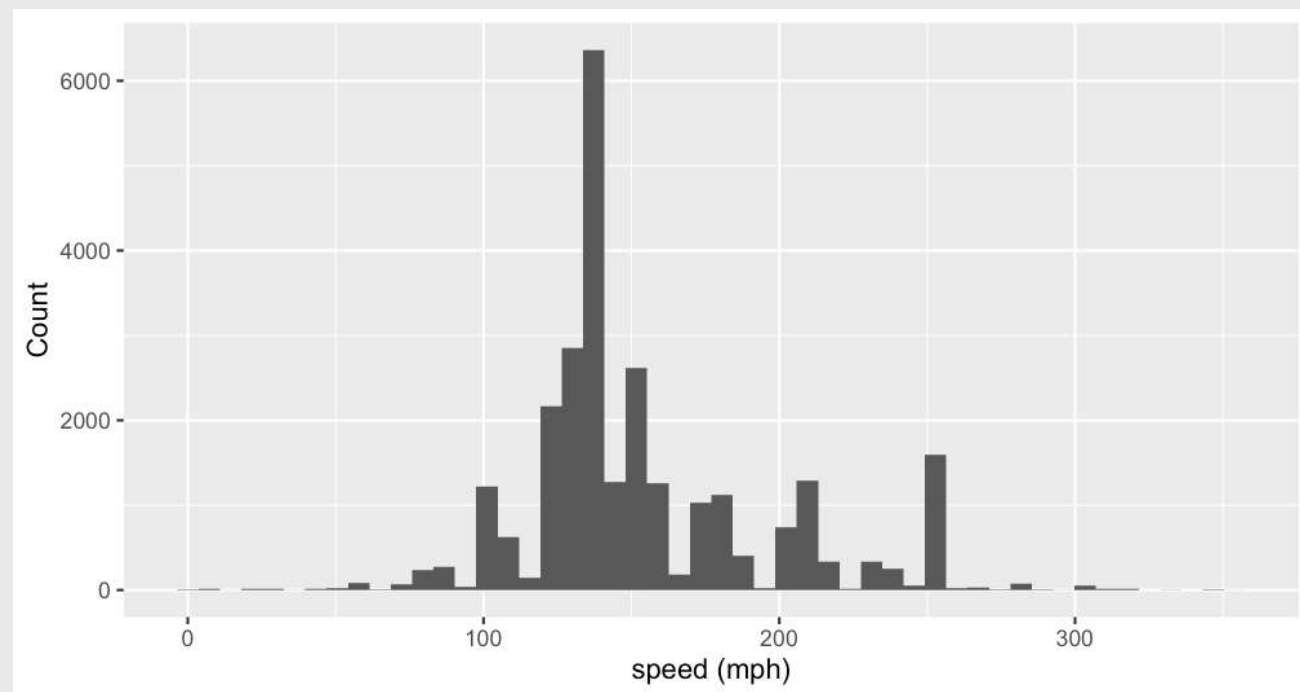
Speed

```
1 ggplot(wildlife_impacts) +  
2   geom_boxplot(aes(x = speed)) +  
3   labs(x = 'Speed (mph)', y = NULL)
```



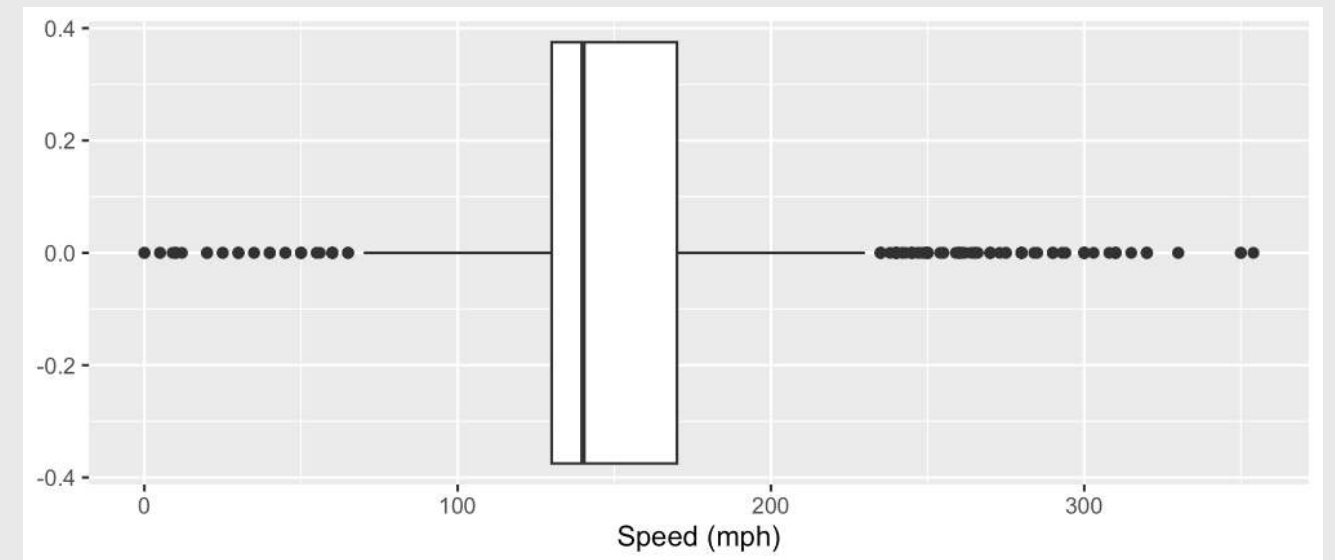
Histogram

- Skewness
- Modes



Boxplot

- Outliers



Practicing visual summaries

15:00

- `faithful.csv`
1. Read in the following data sets: • `marathon.csv`
 2. Summarize the following variables using an appropriate chart (bar chart, histogram, and / or boxplot):
 - faithful: `eruptions`
 - faithful: `waiting`
 - marathon: `Age`
 - marathon: `State`
 - marathon: `Country`
 - marathon: ``Official Time``

Break!

Stand up, Move around, Stretch!

05 : 00

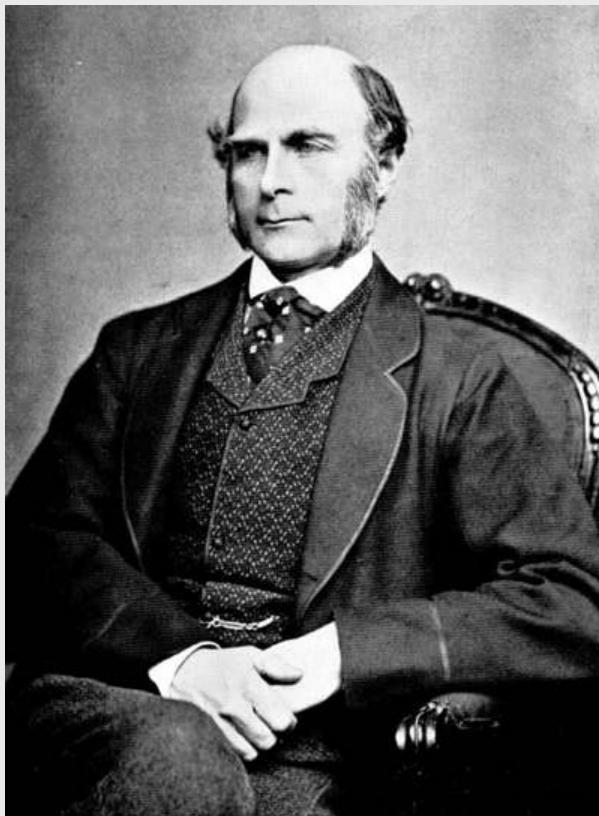
Week 6: *Exploring Data*

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Correlation: a rather racist origin in **eugenics** (“well born”)

Sir Francis Galton (1822 - 1911)

- Charles Darwin's cousin.
- “Father” of **eugenics**.
- Interested in heredity.



Karl Pearson (1857 - 1936)

- Galton's (**hero-worshiping**) protégé.
- Defined correlation equation.
- “Father” of mathematical statistics.



Galton's family data

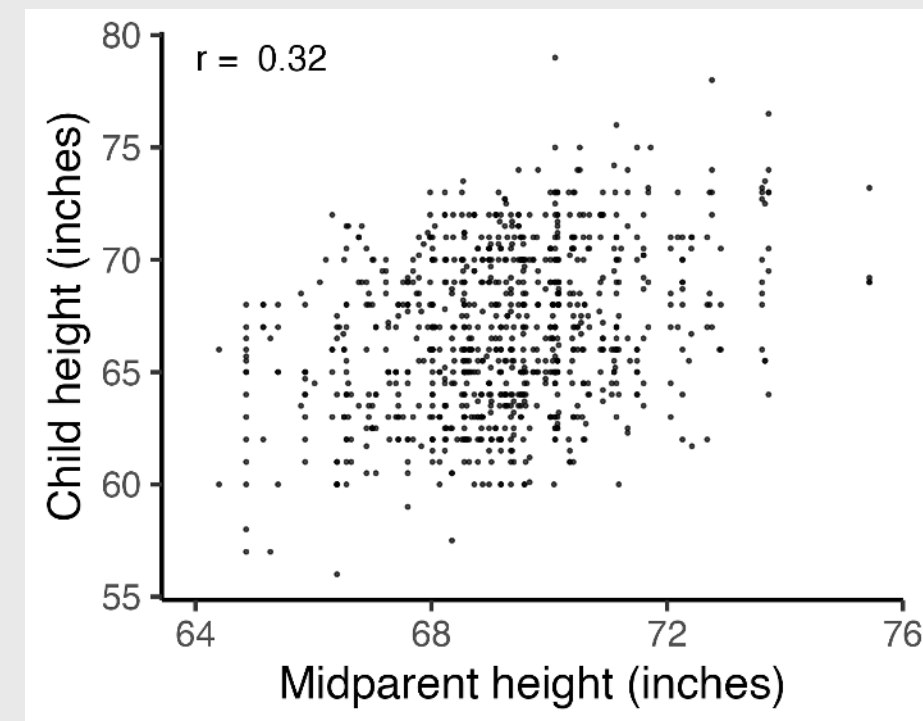
Galton, F. (1886). "Regression towards mediocrity in hereditary stature". *The Journal of the Anthropological Institute of Great Britain and Ireland* 15: 246-263.

Galton's question: Does marriage selection indicate a relationship between the heights of husbands and wives? (He called this "assortative mating")

"midparent height" is just a scaled mean:

```
1 midparentHeight <- (father + 1.08 * mother) /
```

```
1 library(HistData)
2
3 galtonScatterplot <- ggplot(GaltonFamilies) +
4   geom_point(
5     aes(x = midparentHeight, y = childHeight),
6     size = 0.5,
7     alpha = 0.7
8   ) +
9   theme_classic() +
10  labs(
11    x = 'Midparent height (inches)',
12    y = 'Child height (inches)'
13  )
```



How do you measure correlation?

Pearson came up with this:

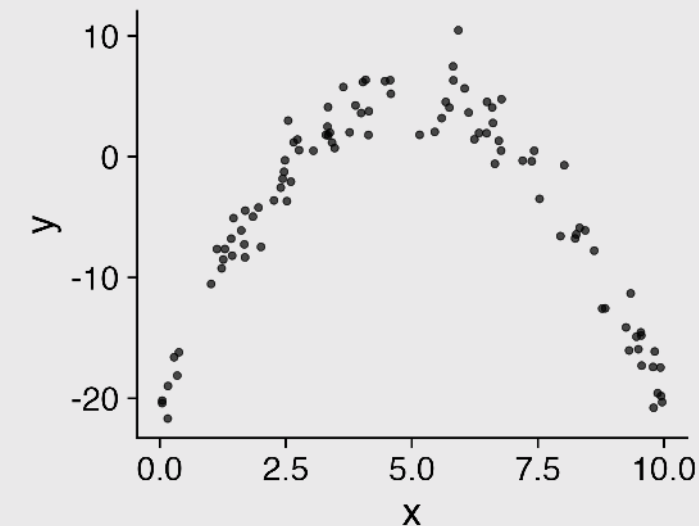
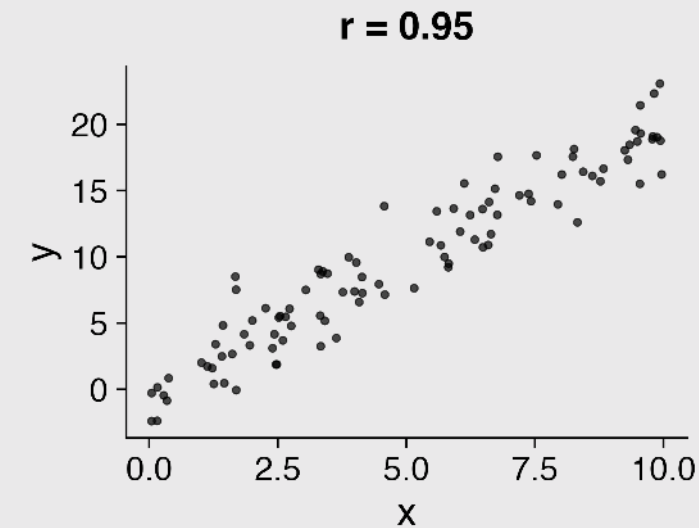
$$r = \frac{\text{Cov}(x,y)}{\text{sd}(x) * \text{sd}(y)}$$

How do you measure correlation?

$$r = \frac{\text{Cov}(x,y)}{\text{sd}(x) * \text{sd}(y)}$$

Assumptions:

1. Variables must be interval or ratio
2. Linear relationship



How do you *interpret* r ?

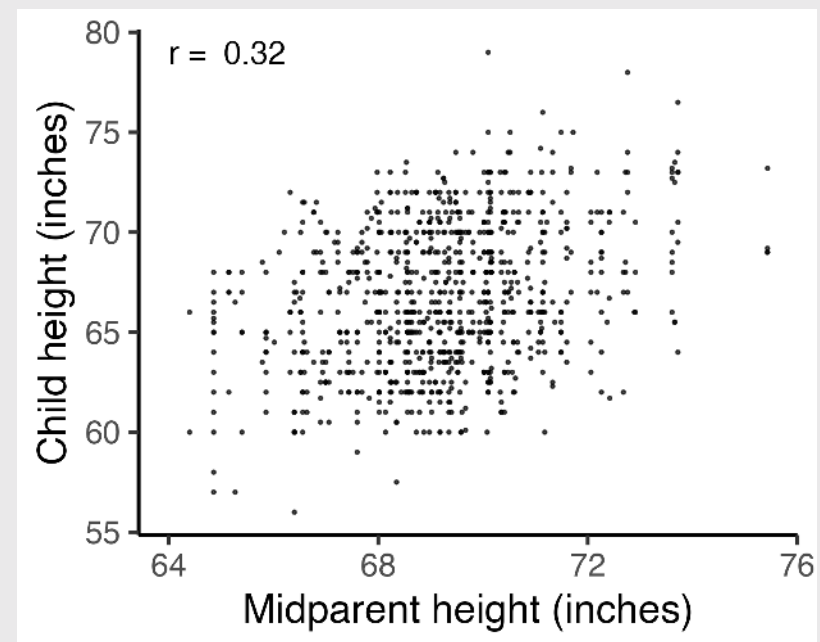
$$r = \frac{\text{Cov}(x,y)}{\text{sd}(x) * \text{sd}(y)}$$

Interpretation:

- $-1 \leq r \leq 1$
- Closer to 1 is stronger correlation
- Closer to 0 is weaker correlation

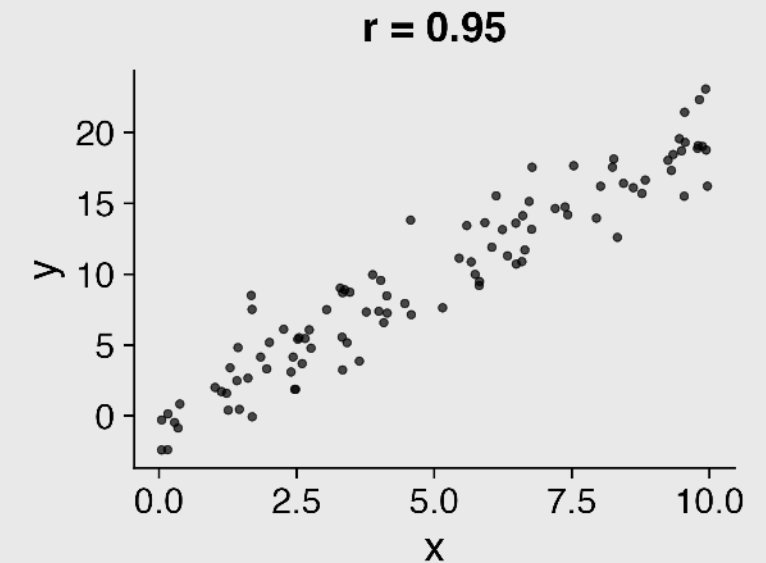
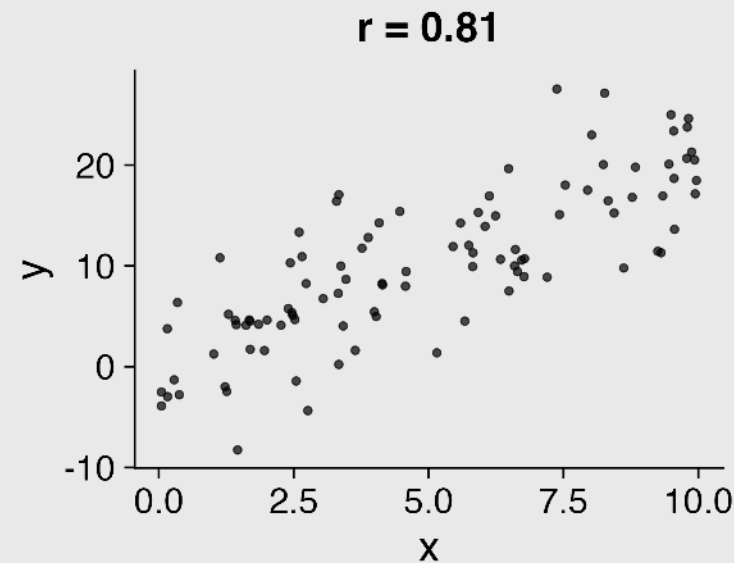
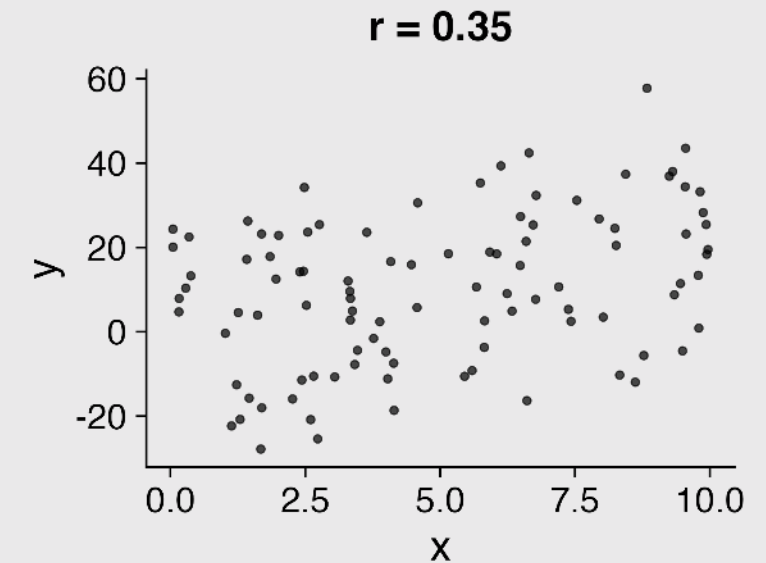
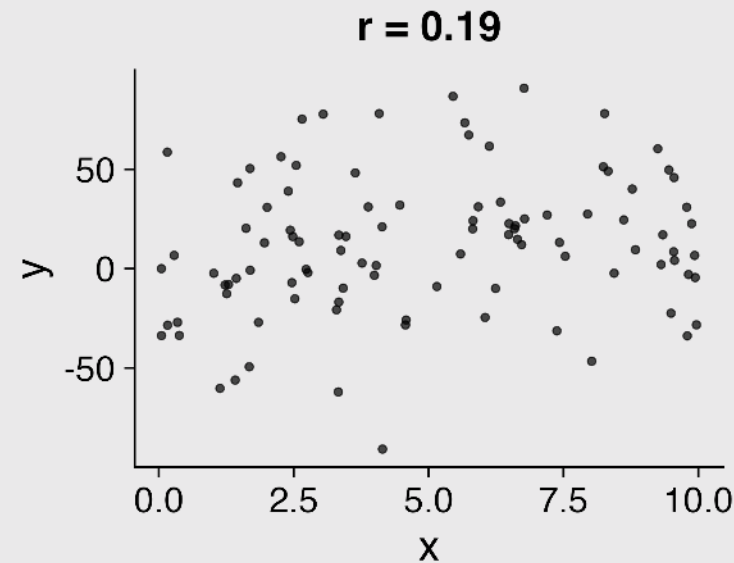
```
1 cor(  
2   x = GaltonFamilies$midparentHeight,  
3   y = GaltonFamilies$childHeight,  
4   method = 'pearson'  
5 )
```

```
#> [1] 0.3209499
```



What does r mean?

- $\pm 0.1 - 0.3$: Weak
- $\pm 0.3 - 0.5$: Moderate
- $\pm 0.5 - 0.8$: Strong
- $\pm 0.8 - 1.0$: Very strong



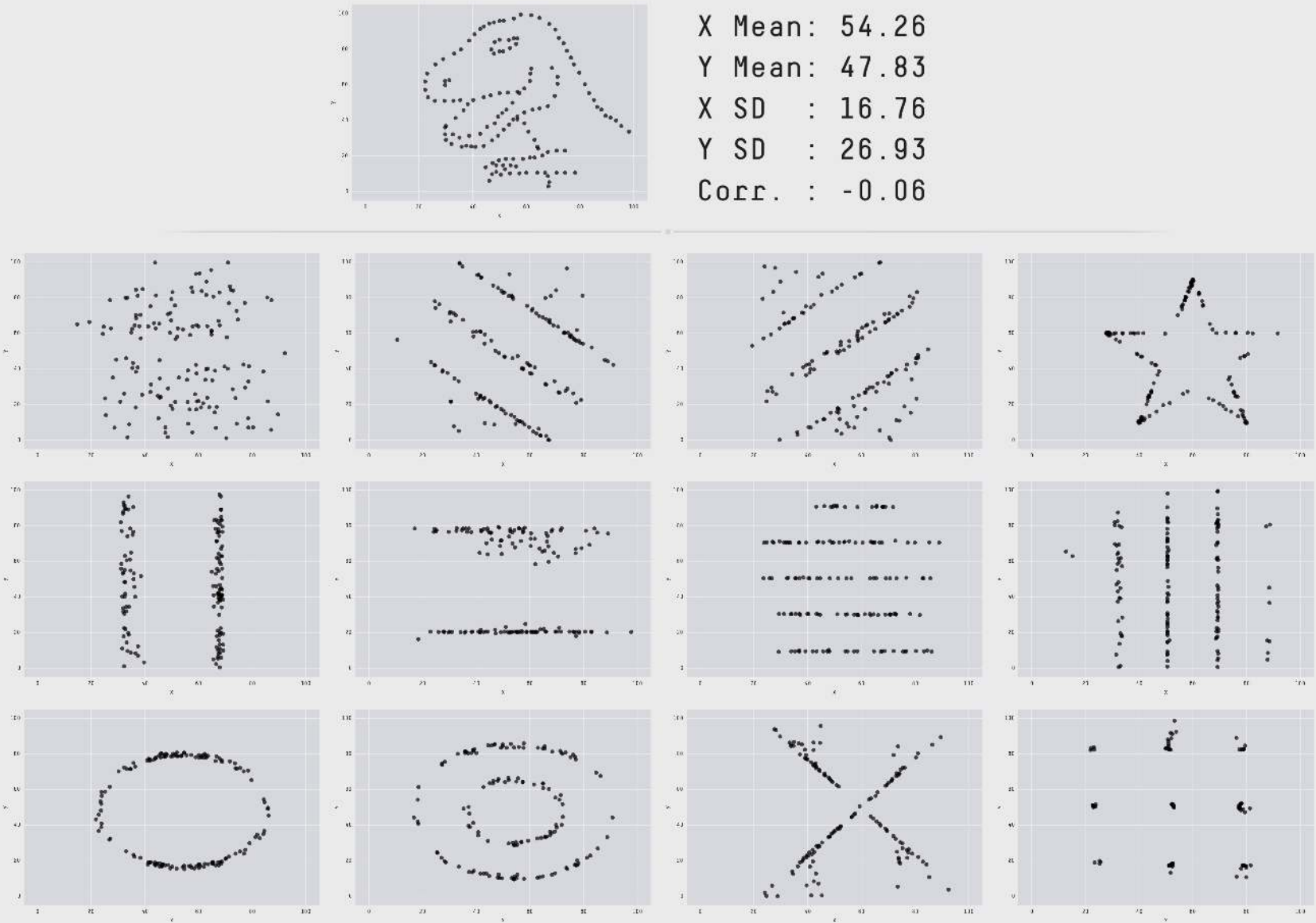
Visualizing correlation is...um...easy, right?

guessthecorrelation.com

Click [here](#) to vote!

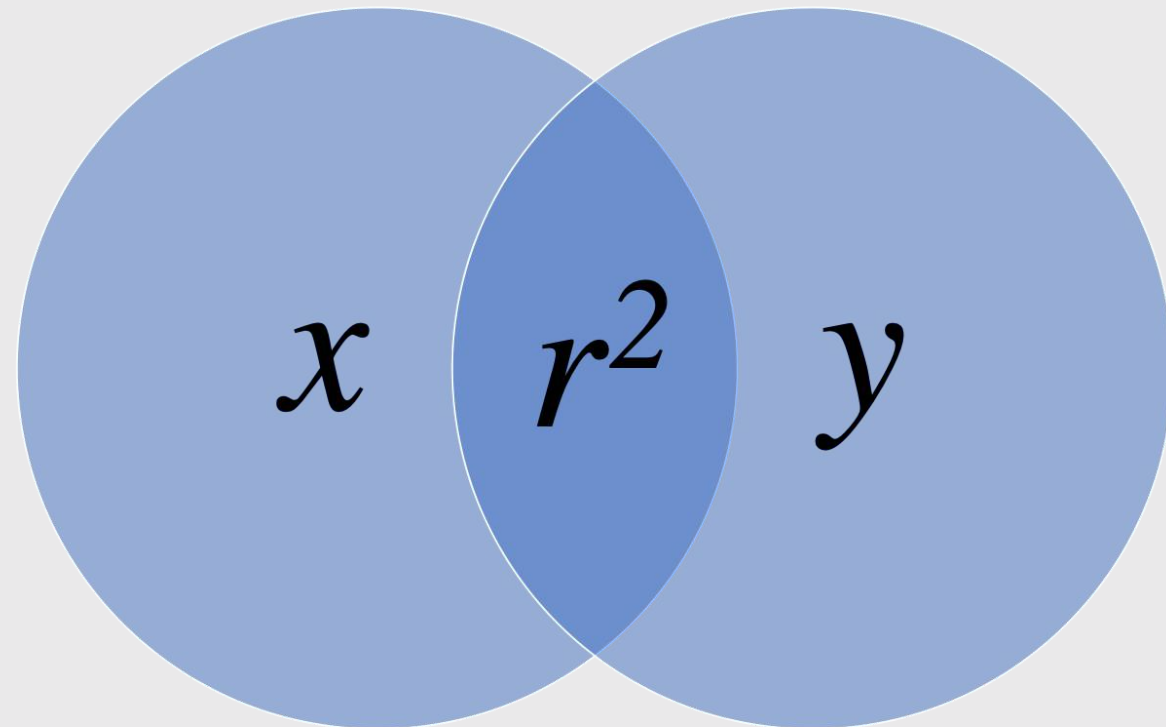
The datasaurus

(More [here](#))



Coefficient of determination: r^2

Percent of variance in one variable that is explained by the other variable



r	r^2
0.1	0.01
0.2	0.04
0.3	0.09
0.4	0.16
0.5	0.25
0.6	0.36
0.7	0.49
0.8	0.64
0.9	0.81
1.0	1.00

You should report both r and r^2

Correlation between parent and child height is 0.32, therefore 10% of the variance in the child height is explained by the parent height.

Correlation != Causation

X causes Y

- Training causes improved performance

Y causes X

- Good (bad) performance causes people to train harder (less hard).

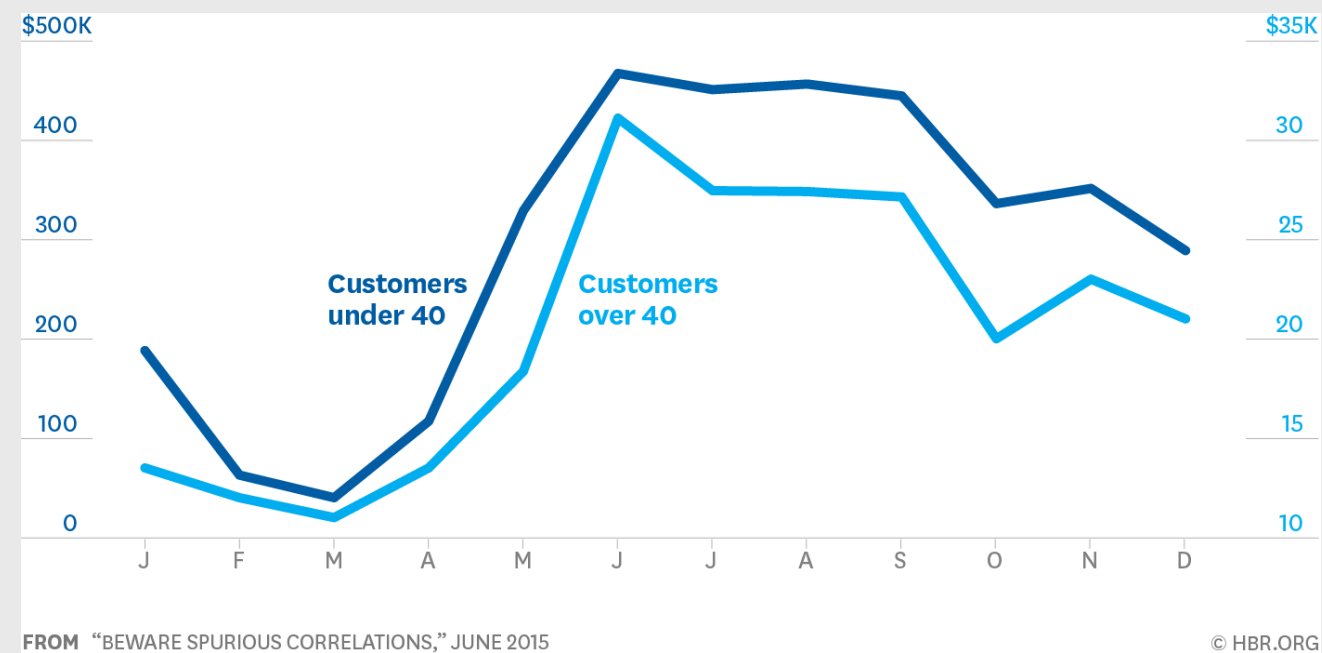
Z causes both X & Y

- Commitment and motivation cause increased training and better performance.

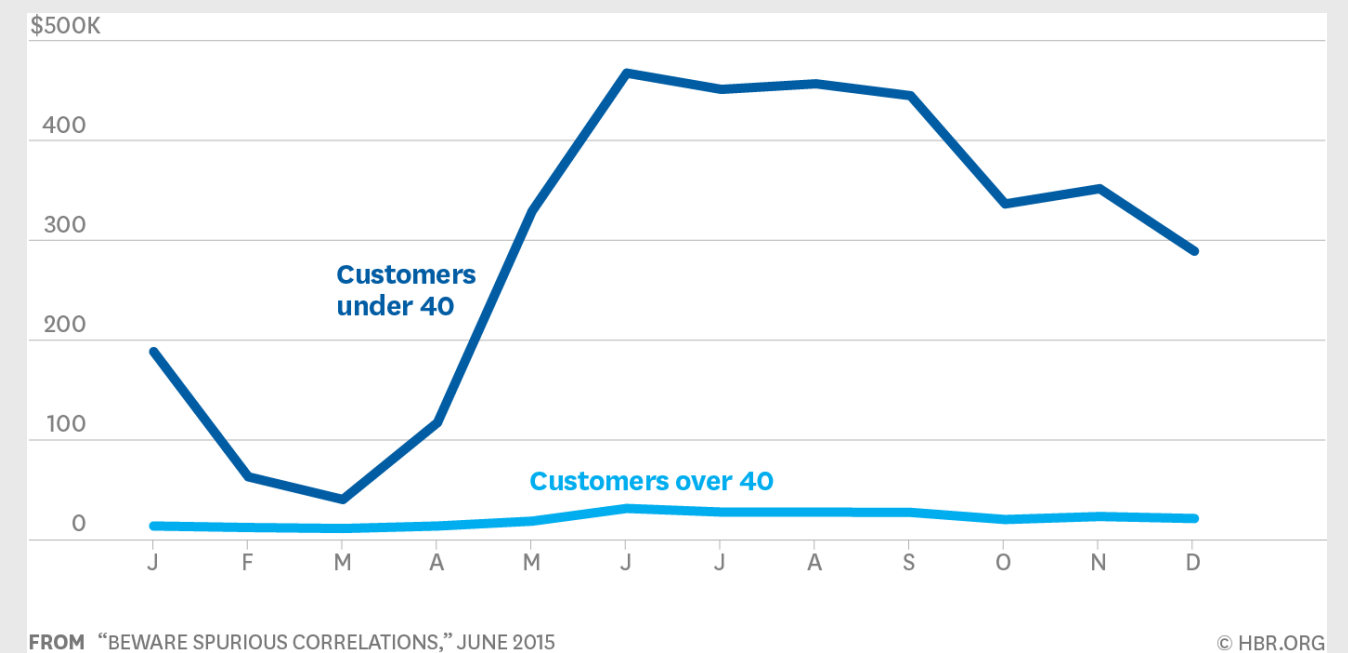
Be weary of dual axes!

(They can cause spurious correlations)

Dual axes

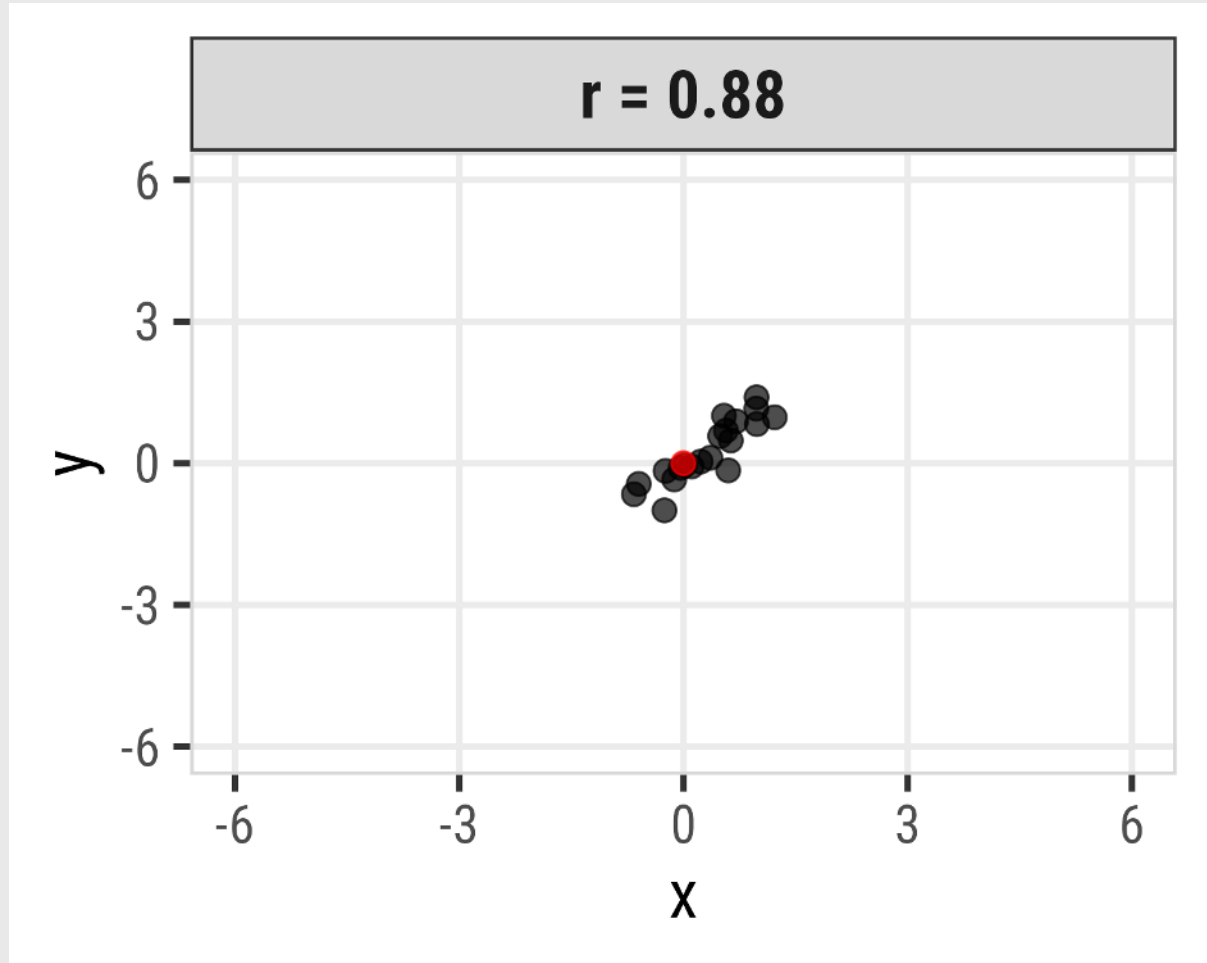


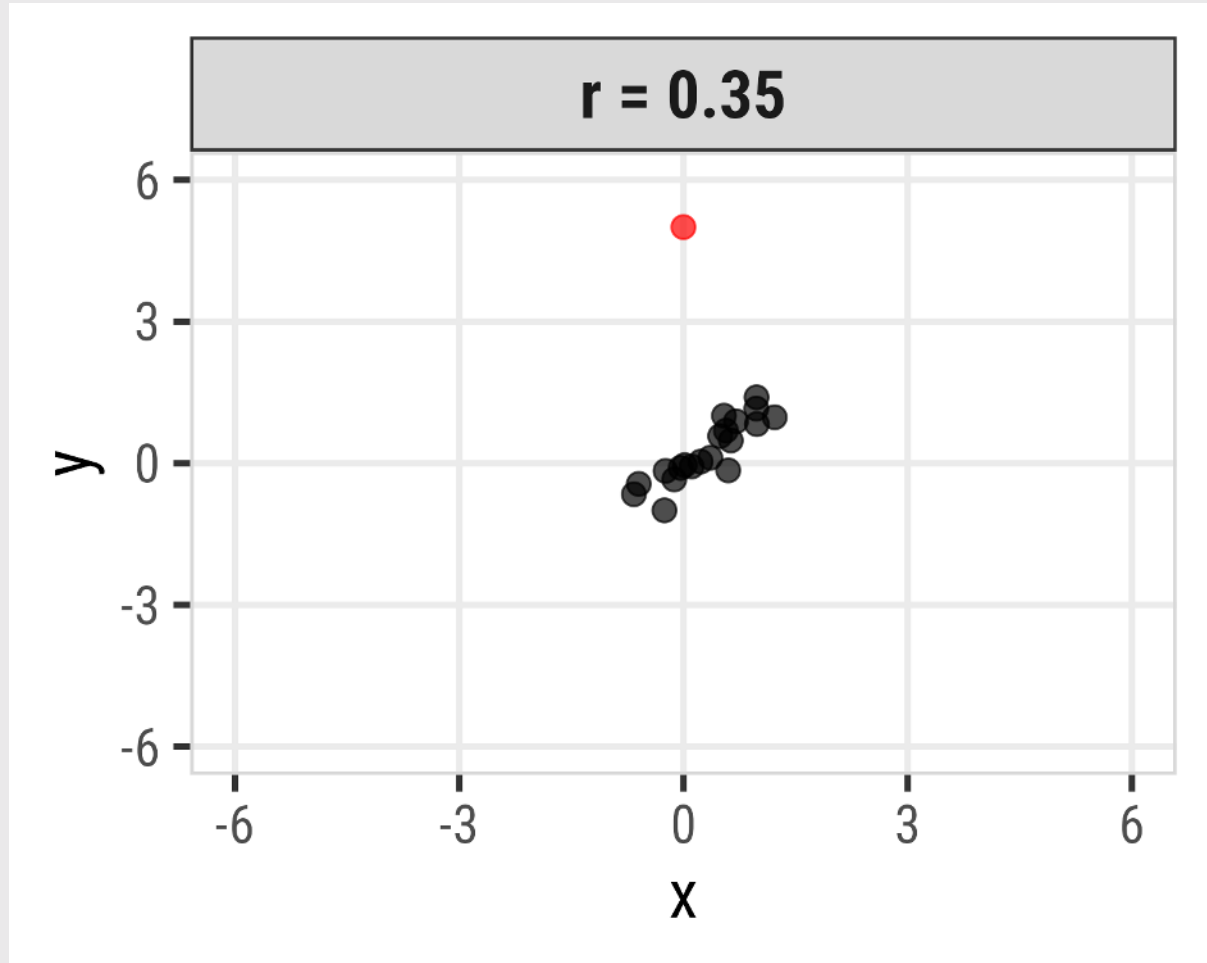
Single axis

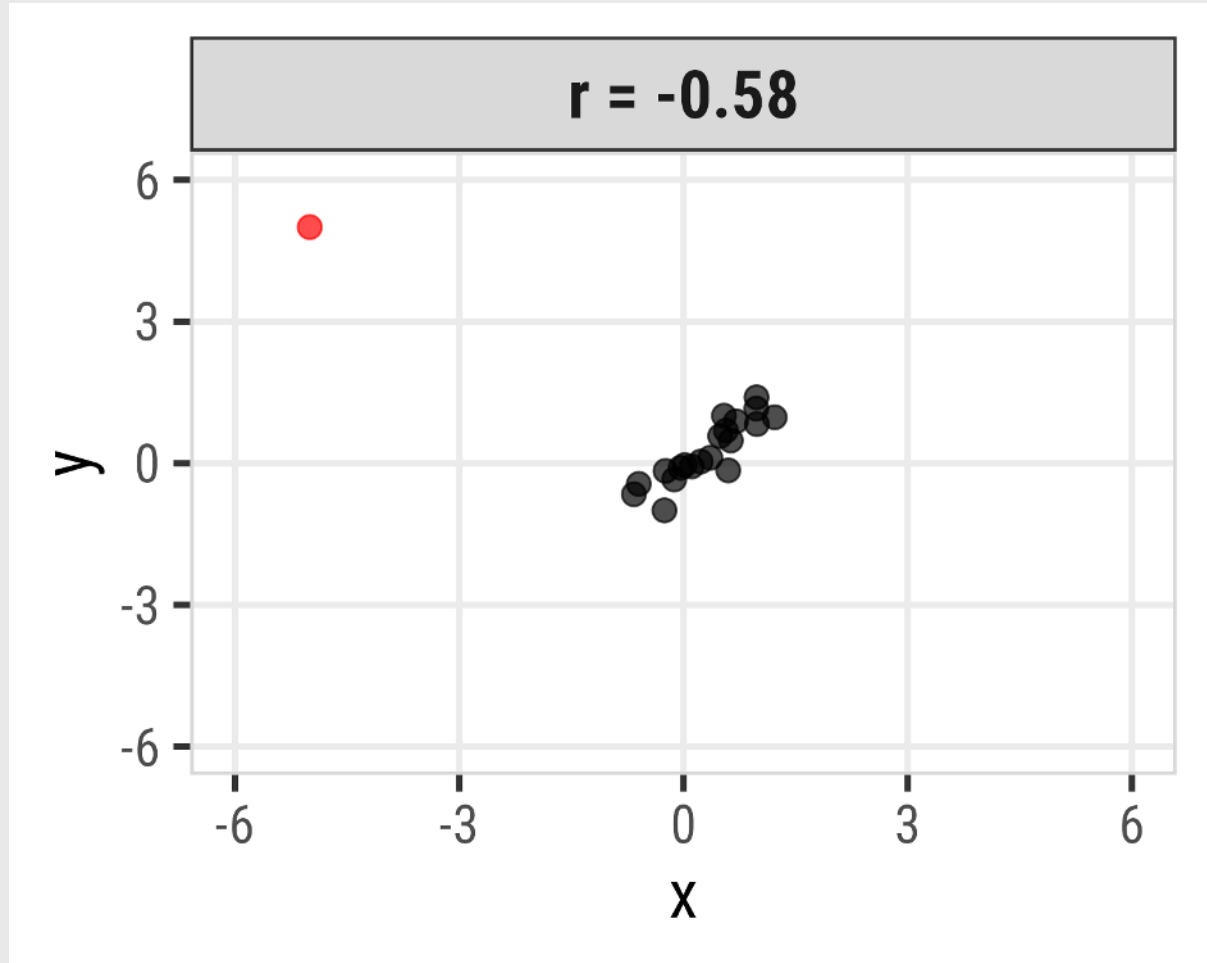


Outliers

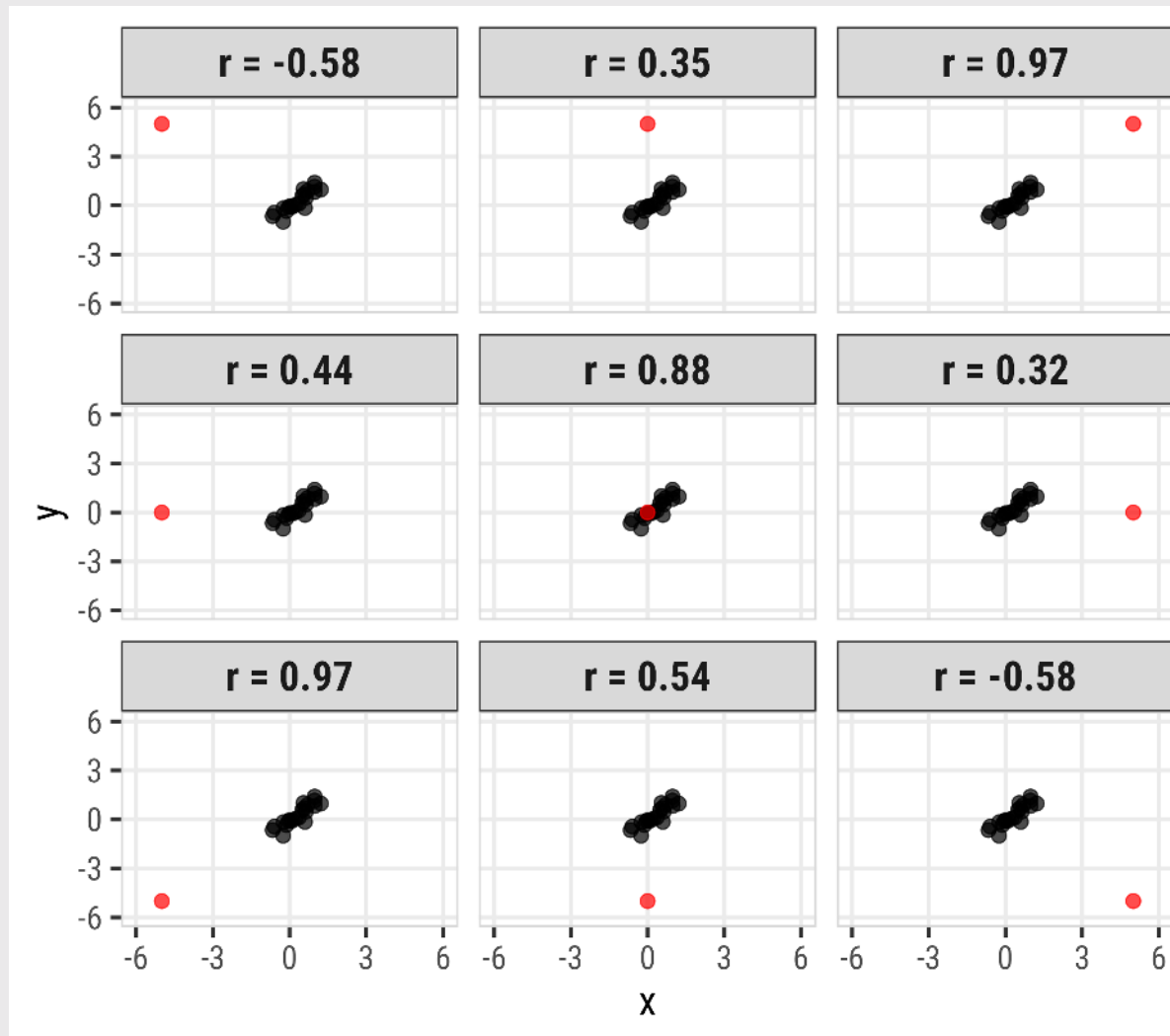








Pearson correlation is highly sensitive to outliers



Spearman's rank-order correlation

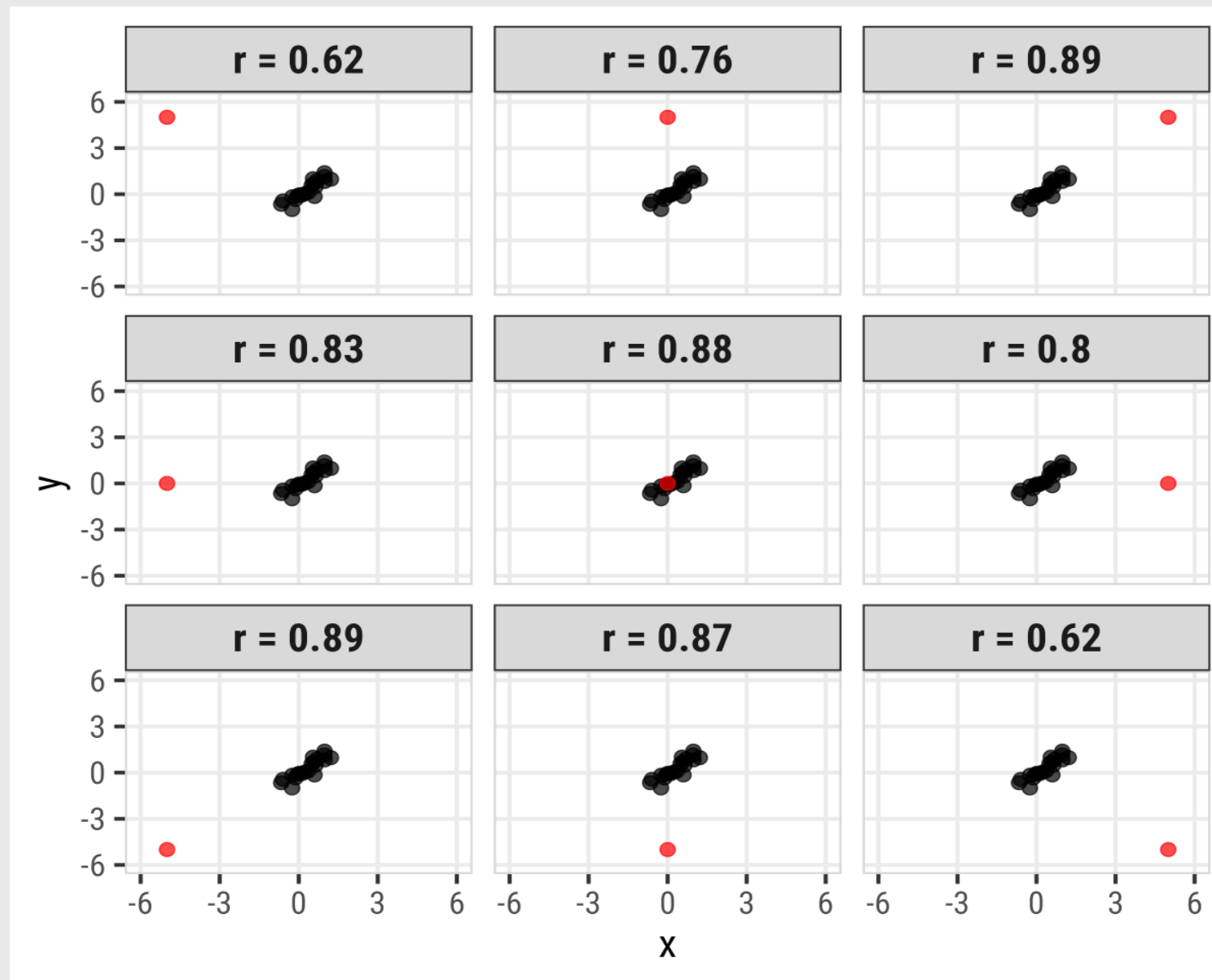
$$r = \frac{\text{Cov}(x,y)}{\text{sd}(x) * \text{sd}(y)}$$

- Separately rank the values of X & Y.
- Use Pearson's correlation on the *ranks* instead of the *x* & *y* values.

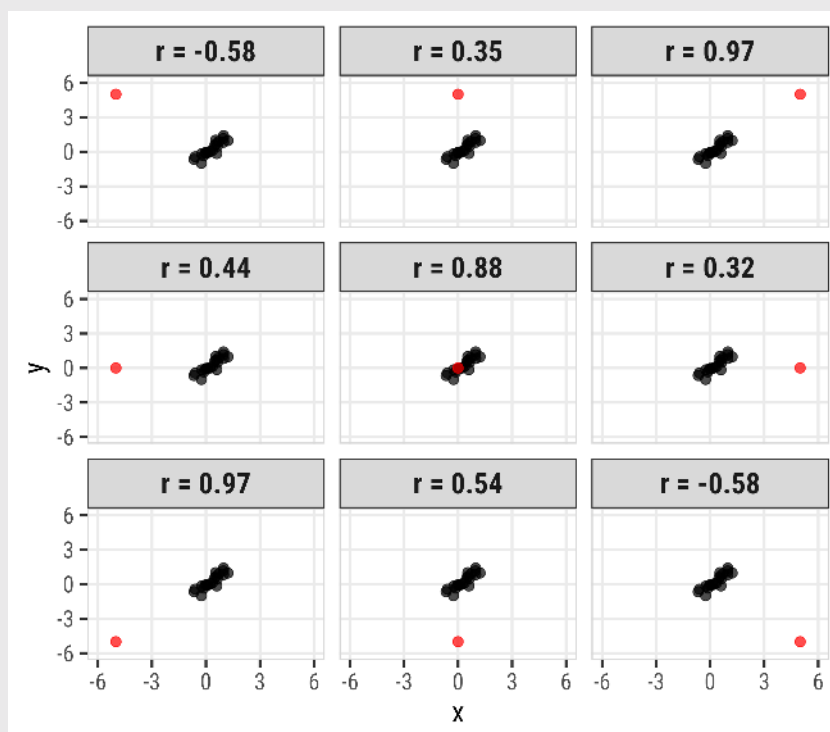
Assumptions:

- Variables can be ordinal, interval or ratio
- Relationship must be monotonic (i.e. does not require linearity)

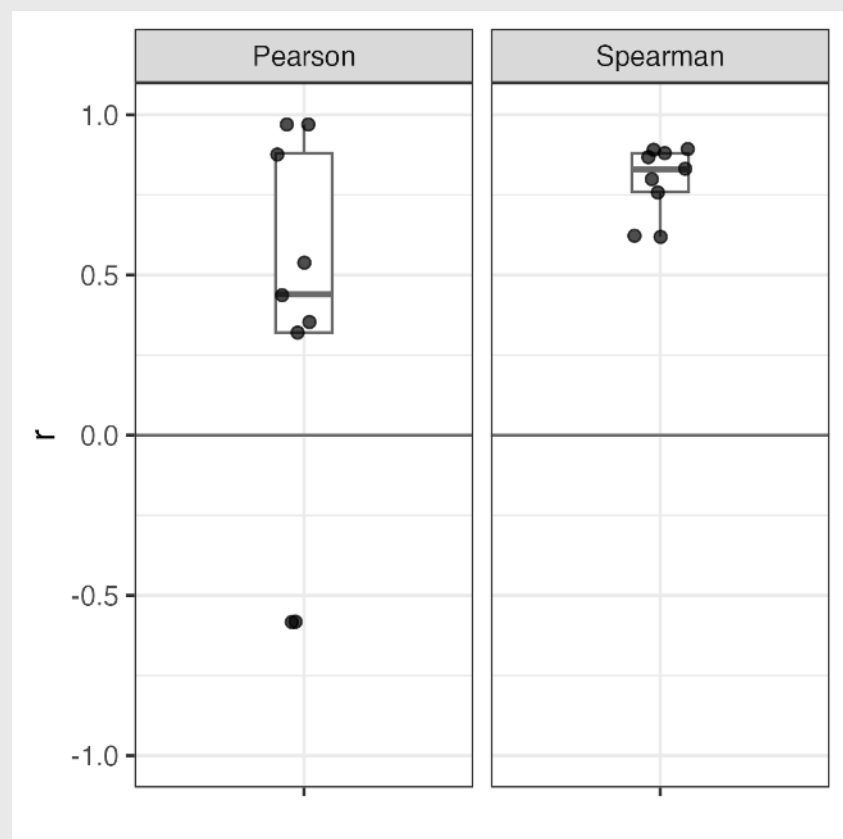
Spearman correlation more robust to outliers



Spearman correlation more robust to outliers



Pearson	Spearman
-0.56	0.53
0.39	0.69
0.94	0.81
0.38	0.76
0.81	0.79
0.31	0.70
0.95	0.81
0.51	0.75
-0.56	0.53



Summary of correlation

- **Pearson's correlation:** Described the strength of a **linear** relationship between two variables that are interval or ratio in nature.
- **Spearman's rank-order correlation:** Describes the strength of a **monotonic** relationship between two variables that are ordinal, interval, or ratio. **It is more robust to outliers.**
- The **coefficient of determination** (r^2) describes the amount of variance in one variable that is explained by the other variable.
- **Correlation != Causation**

R command (hint: add `use = "complete.obs"` to drop NA values)

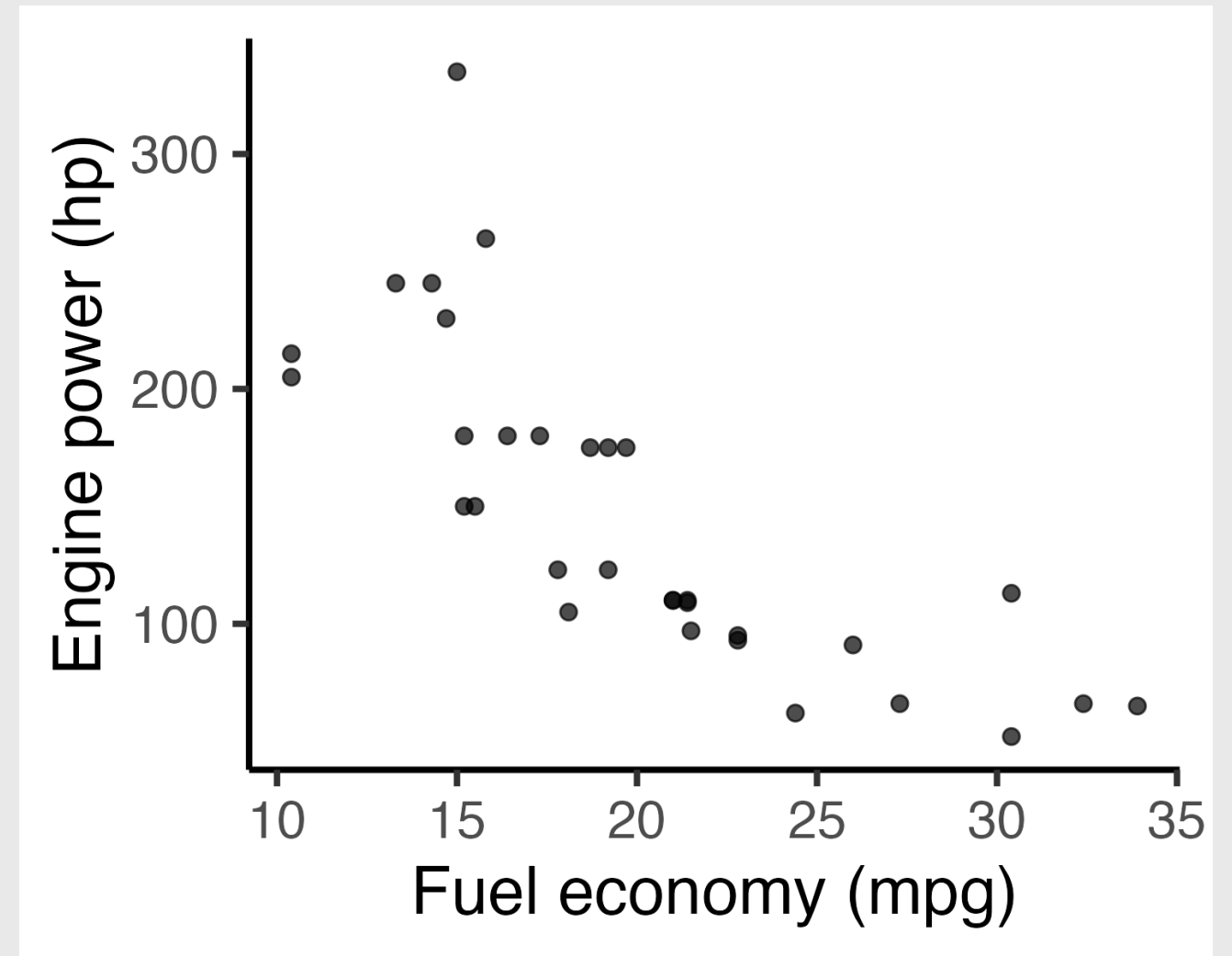
```
1 pearson <- cor(x, y, method = "pearson", use = "complete.obs")  
2 spearman <- cor(x, y, method = "spearman", use = "complete.obs")
```


Week 6: *Exploring Data*

1. Exploring Data
2. Data Types
3. Centrality & Variability
4. Visualizing Centrality & Variability
5. Correlation
6. Visualizing Correlation
7. Visualizing Relationships

Scatterplots: The correlation workhorse

```
1 scatterplot <- mtcars %>%
2   ggplot() +
3   geom_point(
4     aes(x = mpg, y = hp),
5     size = 2,
6     alpha = 0.7
7   ) +
8   theme_classic(base_size = 20) +
9   labs(
10    x = 'Fuel economy (mpg)',
11    y = 'Engine power (hp)'
12  )
13
14 scatterplot
```



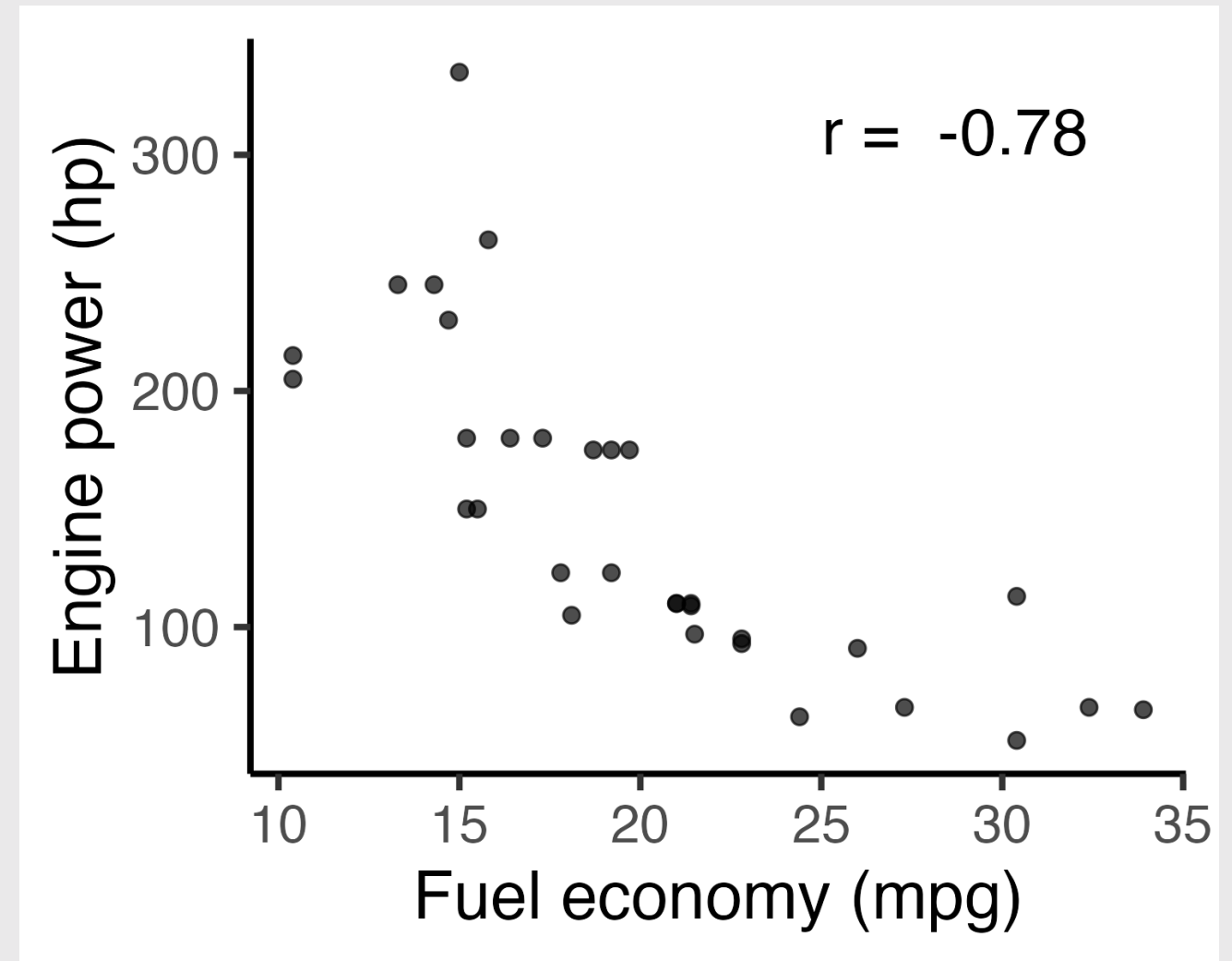
Adding a correlation label to a chart

Make the correlation label

```
1 corr <- cor(  
2   mtcars$mpg,  
3   mtcars$hp,  
4   method = 'pearson'  
5 )  
6  
7 corrLabel <- paste('r = ', round(corr, 2))
```

Add label to the chart with `annotate()`

```
1 scatterplot +  
2   annotate(  
3     geom = 'text',  
4     x = 25,  
5     y = 310,  
6     label = corrLabel,  
7     hjust = 0,  
8     size = 7  
9   )
```



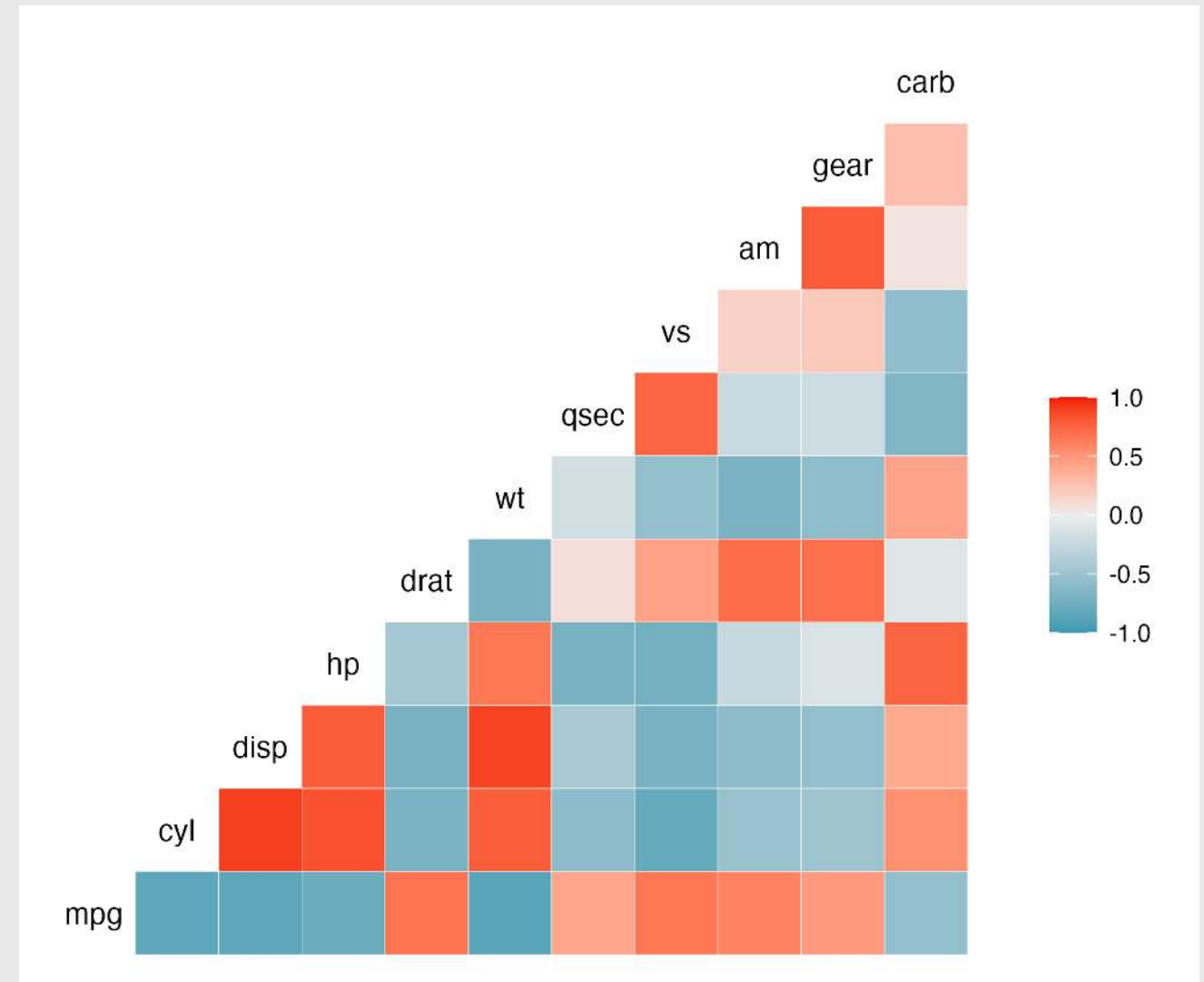
Visualize all the correlations



Visualize all the correlations: `ggcorr()`

```
1 library('GGally')
```

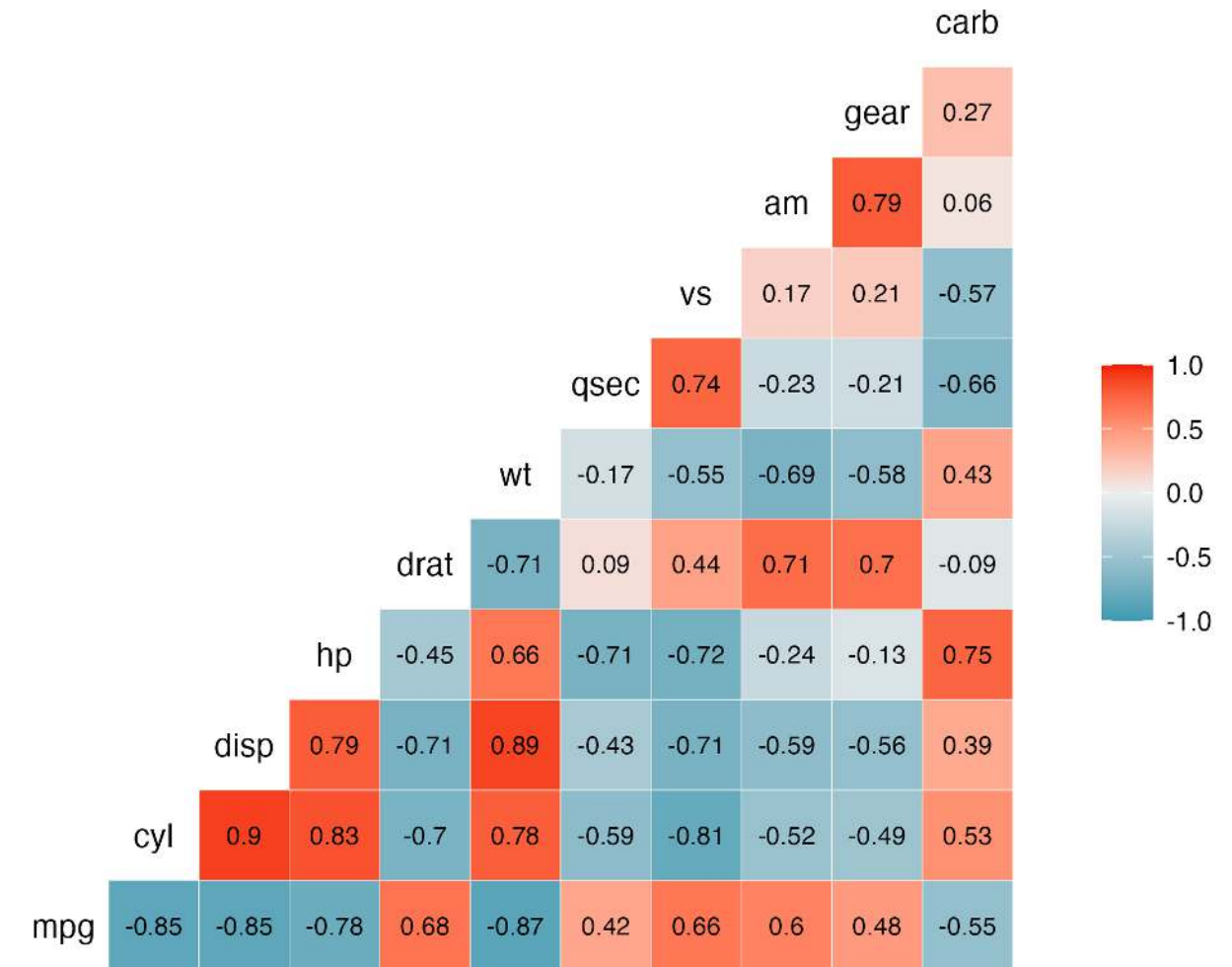
```
1 mtcars %>%  
2   ggcorr()
```



Visualizing correlations: `ggcorr()`

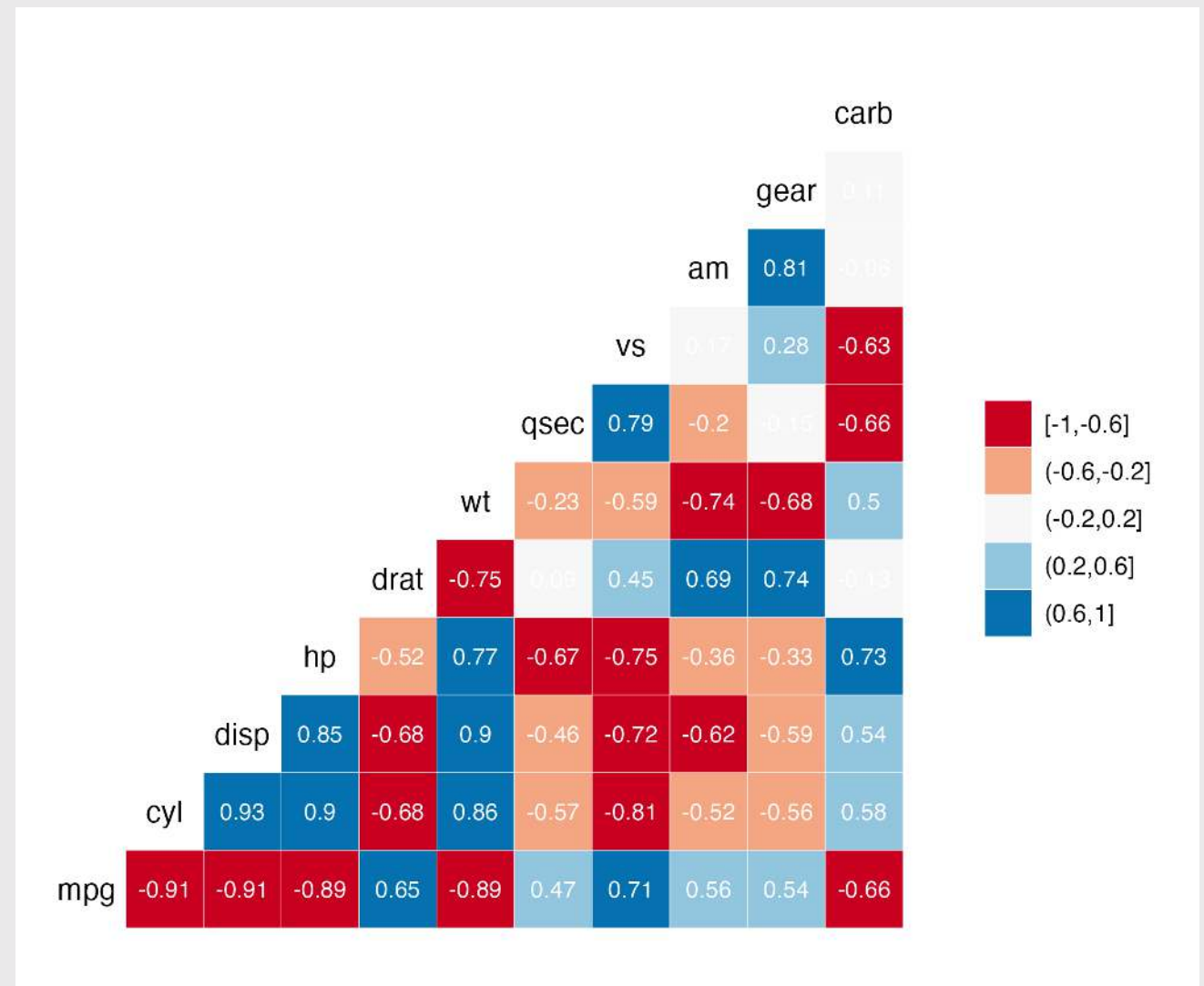
```
1 library('GGally')
```

```
1 mtcars %>%  
2   ggcorr(  
3     label = TRUE,  
4     label_size = 3,  
5     label_round = 2  
6   )
```



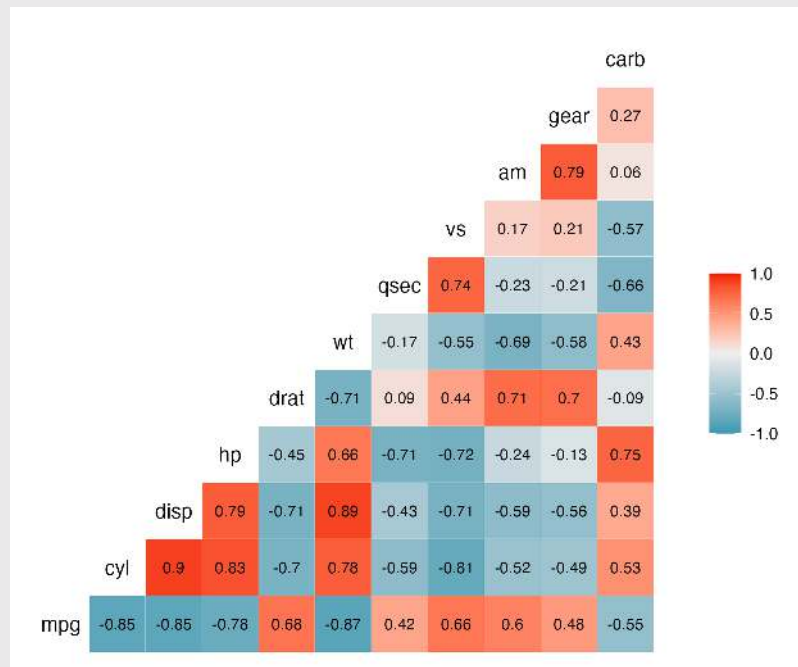
Visualizing correlations: `ggcorr()`

```
1 ggcor_mtcars_final <- mtcars %>%  
2   ggcorr(  
3     label = TRUE,  
4     label_size = 3,  
5     label_round = 2,  
6     label_color = 'white',  
7     nbreaks = 5,  
8     palette = "RdBu"  
9   )
```



Pearson

```
1 mtcars %>%
2   ggcorr(
3     label = TRUE,
4     label_size = 3,
5     label_round = 2,
6     method = c("pairwise", "pearson")
7   )
```



Spearman

```
1 mtcars %>%
2   ggcorr(
3     label = TRUE,
4     label_size = 3,
5     label_round = 2,
6     method = c("pairwise", "spearman")
7   )
```

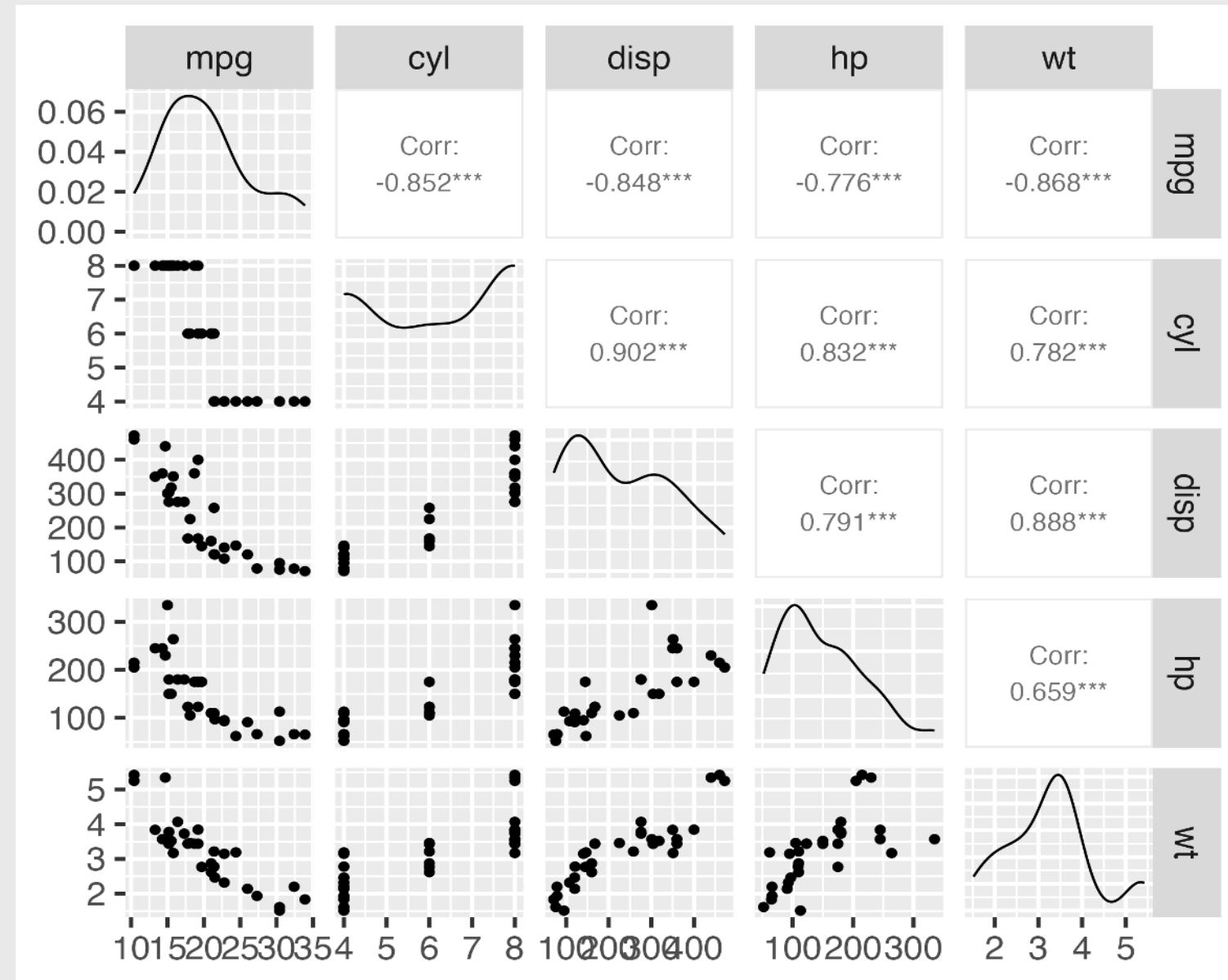


Correlograms: `ggpairs()`

```
1 library('GGally')
```

```
1 mtcars %>%  
2   select(mpg, cyl, disp, hp, wt) %>%  
3   ggpairs()
```

- Look for linear relationships
- View distribution of each variable

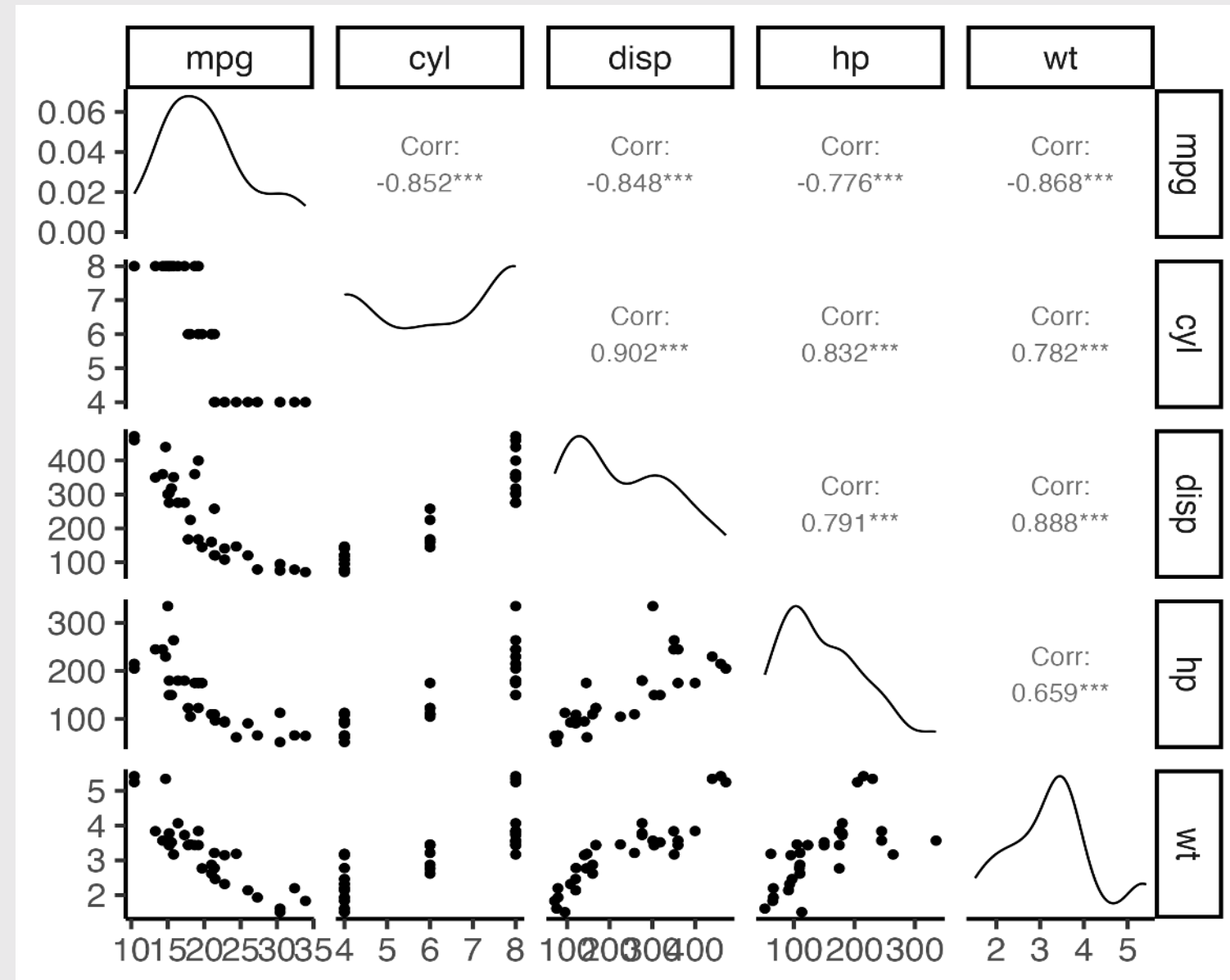


Correlograms: `ggpairs()`

```
1 library('GGally')
```

```
1 mtcars %>%  
2   select(mpg, cyl, disp, hp, wt) %>%  
3   ggpairs() +  
4   theme_classic()
```

- Look for linear relationships
- View distribution of each variable



Your turn

15:00

Using the `penguins` data frame:

1. Find the two variables with the largest correlation in absolute value (i.e. closest to -1 or 1).
2. Create a scatter plot of those two variables.
3. Add an annotation for the Pearson correlation coefficient.

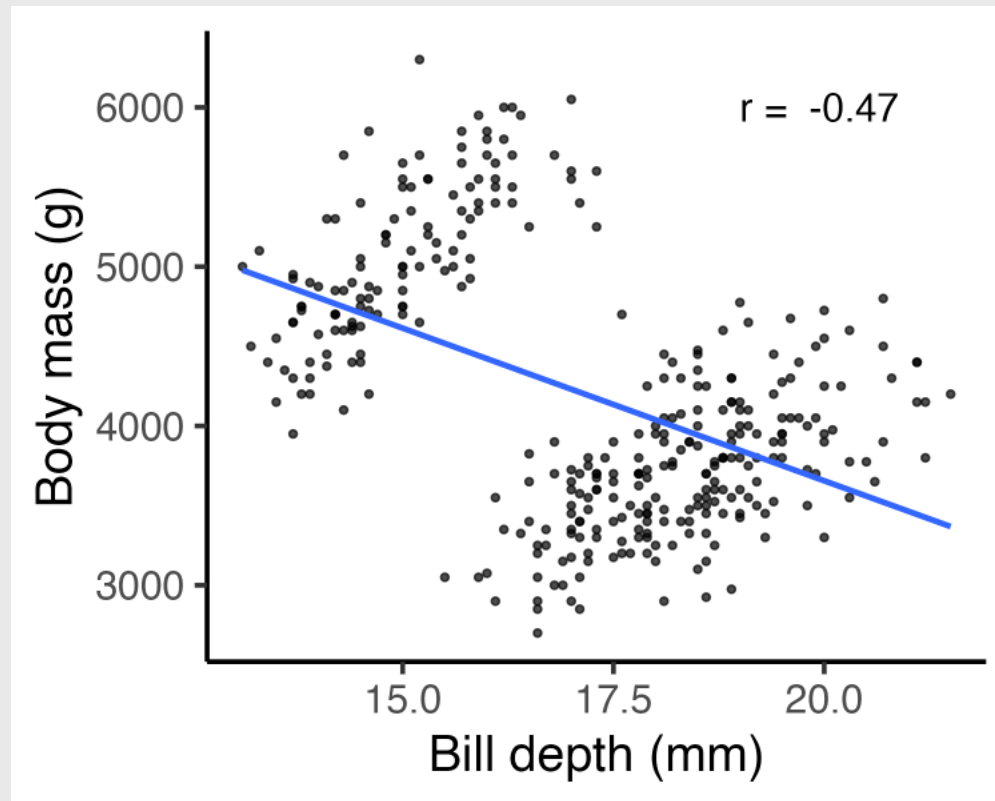
[palmerpenguins library](#)



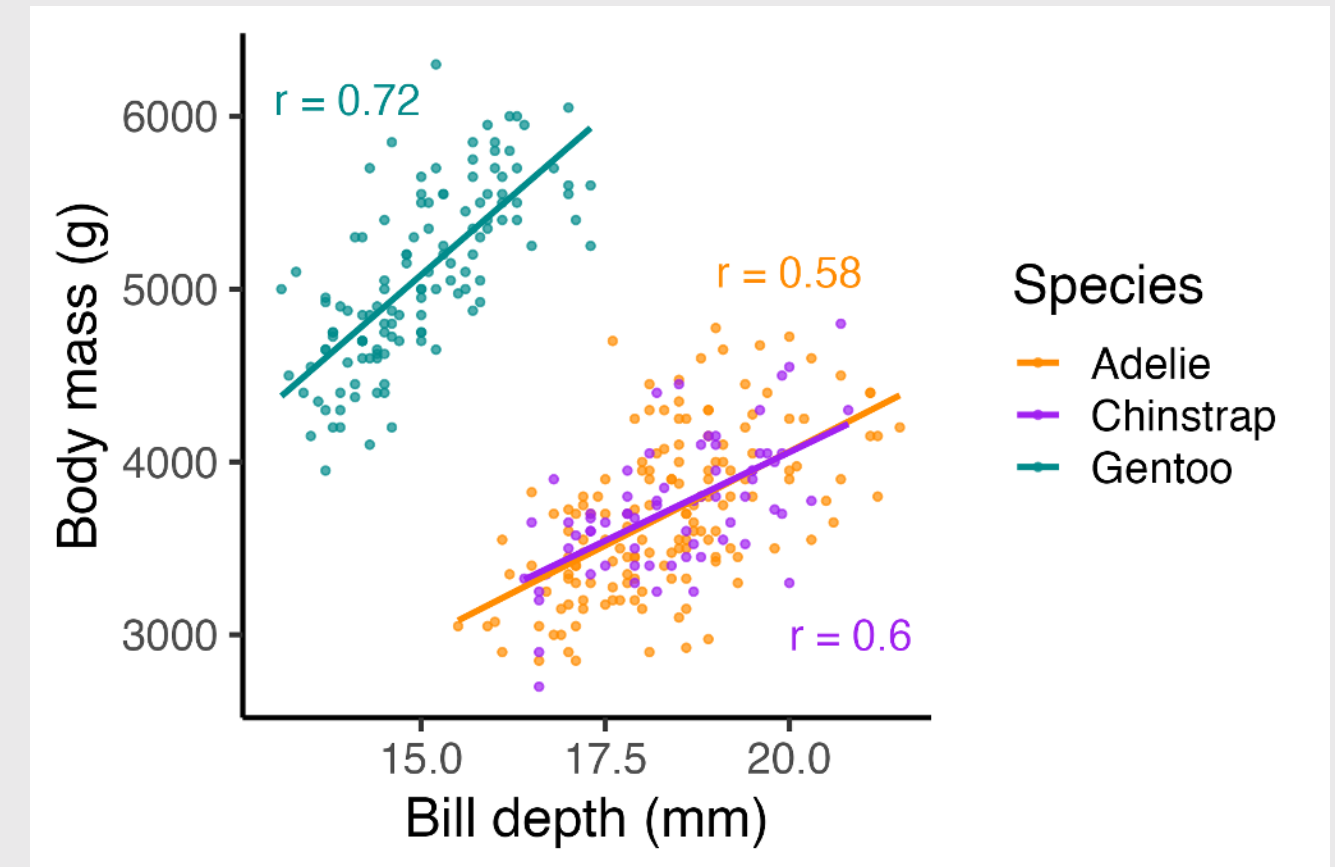
Artwork by [@allison_horst](#)

Simpson's Paradox: when correlation betrays you

Body mass vs. Bill depth



Body mass vs. Bill depth



Week 6: *Exploring Data*

1. Exploring Data
2. Data Types
3. Centrality & Variability
4. Visualizing Centrality & Variability
5. Correlation
6. Visualizing Correlation
7. Visualizing Relationships

Visualizing variation

Ask yourself:

- What type of **variation** occurs within my variables?
- What type of **covariation** occurs between my variables?

Check out [these guides](#)

Variation		Covariation	
		Categorical Y	Continuous Y
Categorical X	Categorical	Heatmap or Count	Boxplot
	Continuous	Boxplot (with coord_flip)	Scatterplot (many to one) line chart (one to one)

Two Categorical Variables

Summarize with a table of counts

```
1 wildlife_impacts %>%  
2   count(operator, time_of_day)
```

```
#> # A tibble: 20 × 3  
#>   operator      time_of_day      n  
#>   <chr>      <chr>      <int>  
#> 1 AMERICAN AIRLINES Dawn         458  
#> 2 AMERICAN AIRLINES Day          7809  
#> 3 AMERICAN AIRLINES Dusk          584  
#> 4 AMERICAN AIRLINES Night         3710  
#> 5 AMERICAN AIRLINES <NA>         2326  
#> 6 DELTA AIR LINES Dawn           267  
#> 7 DELTA AIR LINES Day          4846  
#> 8 DELTA AIR LINES Dusk           353  
#> 9 DELTA AIR LINES Night         2090  
#> 10 DELTA AIR LINES <NA>         1449
```


Two Categorical Variables

Convert to "wide" format with `pivot_wider()` to make it easier to compare values

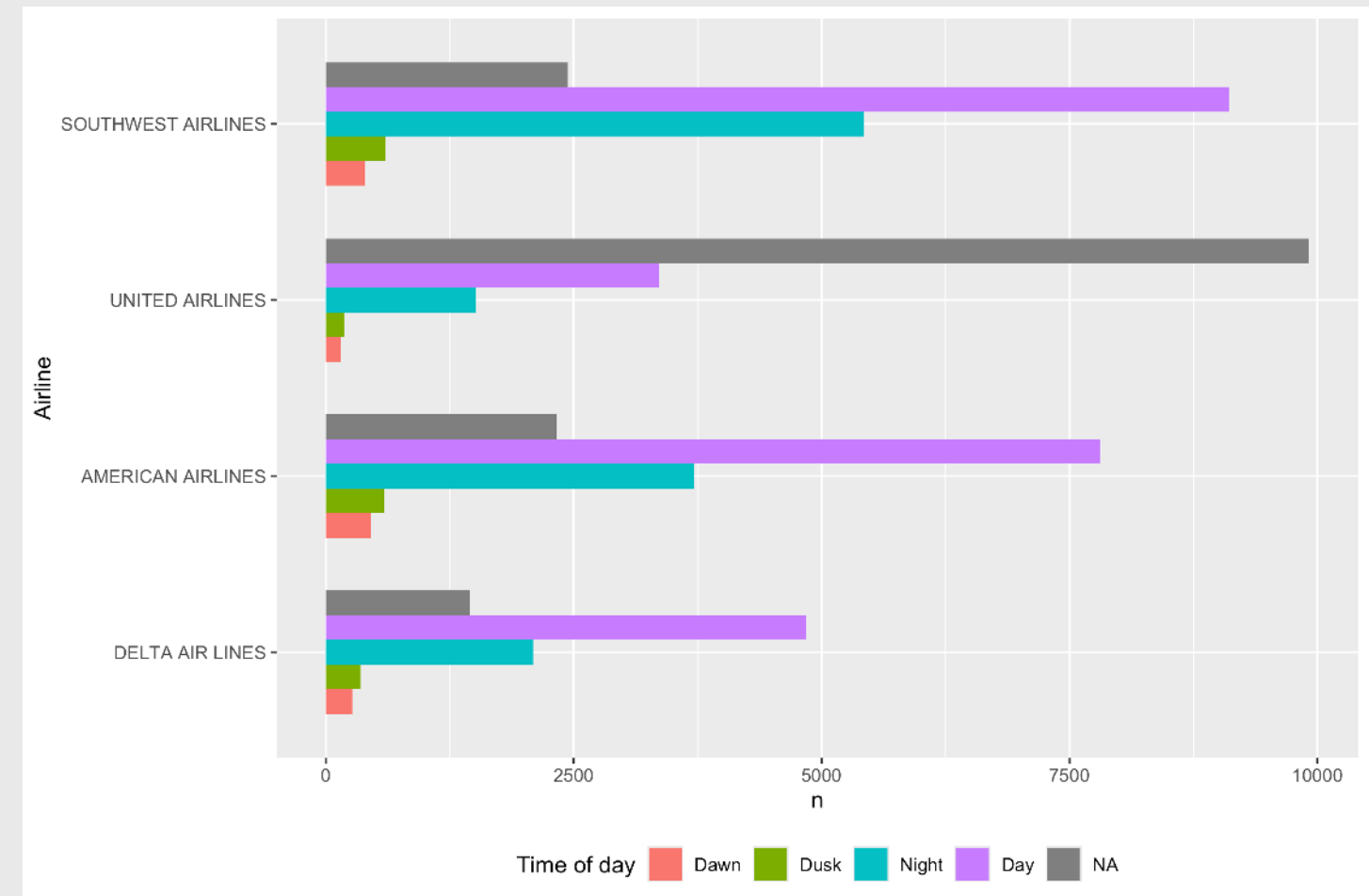
```
1 wildlife_impacts %>%  
2   count(operator, time_of_day) %>%  
3   pivot_wider(names_from = time_of_day, values_from = n)
```

```
#> # A tibble: 4 × 6  
#>   operator      Dawn    Day    Dusk  Night  `NA`  
#>   <chr>      <int> <int> <int> <int> <int>  
#> 1 AMERICAN AIRLINES    458  7809   584  3710  2326  
#> 2 DELTA AIR LINES     267  4846   353  2090  1449  
#> 3 SOUTHWEST AIRLINES   394  9109   599  5425  2443  
#> 4 UNITED AIRLINES     151  3359   181  1510  9915
```


Two Categorical Variables

Visualize with bars:
map **fill** to denote 2nd categorical var

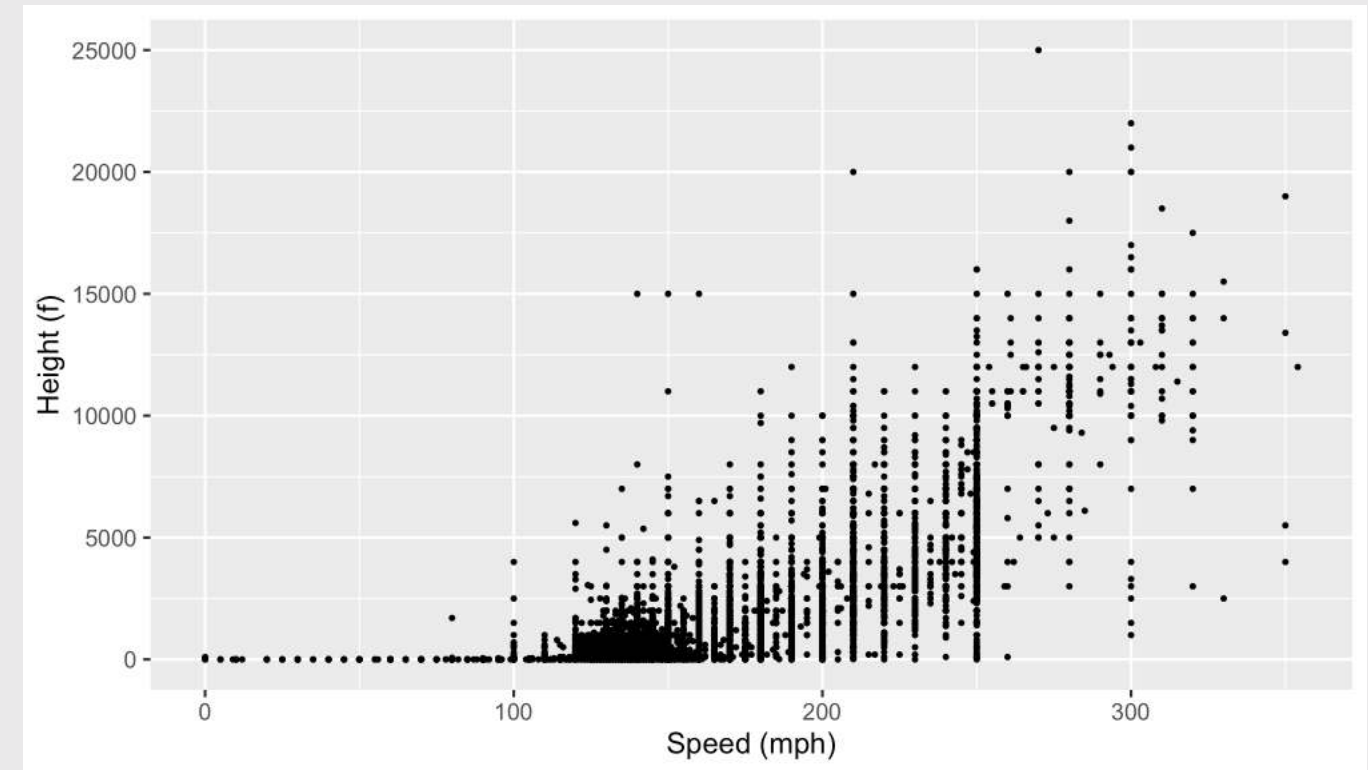
```
1 wildlife_impacts %>%
2   count(operator, time_of_day) %>%
3   ggplot() +
4   geom_col(
5     aes(
6       x = n,
7       y = reorder(operator, n),
8       fill = reorder(time_of_day, n)
9     ),
10    width = 0.7,
11    position = 'dodge'
12  ) +
13  theme(legend.position = "bottom") +
14  labs(
15    fill = "Time of day",
16    y = "Airline"
17  )
```



Two Continuous Variables

Visualize with scatterplot - looking for *clustering* and/or *correlational* relationship

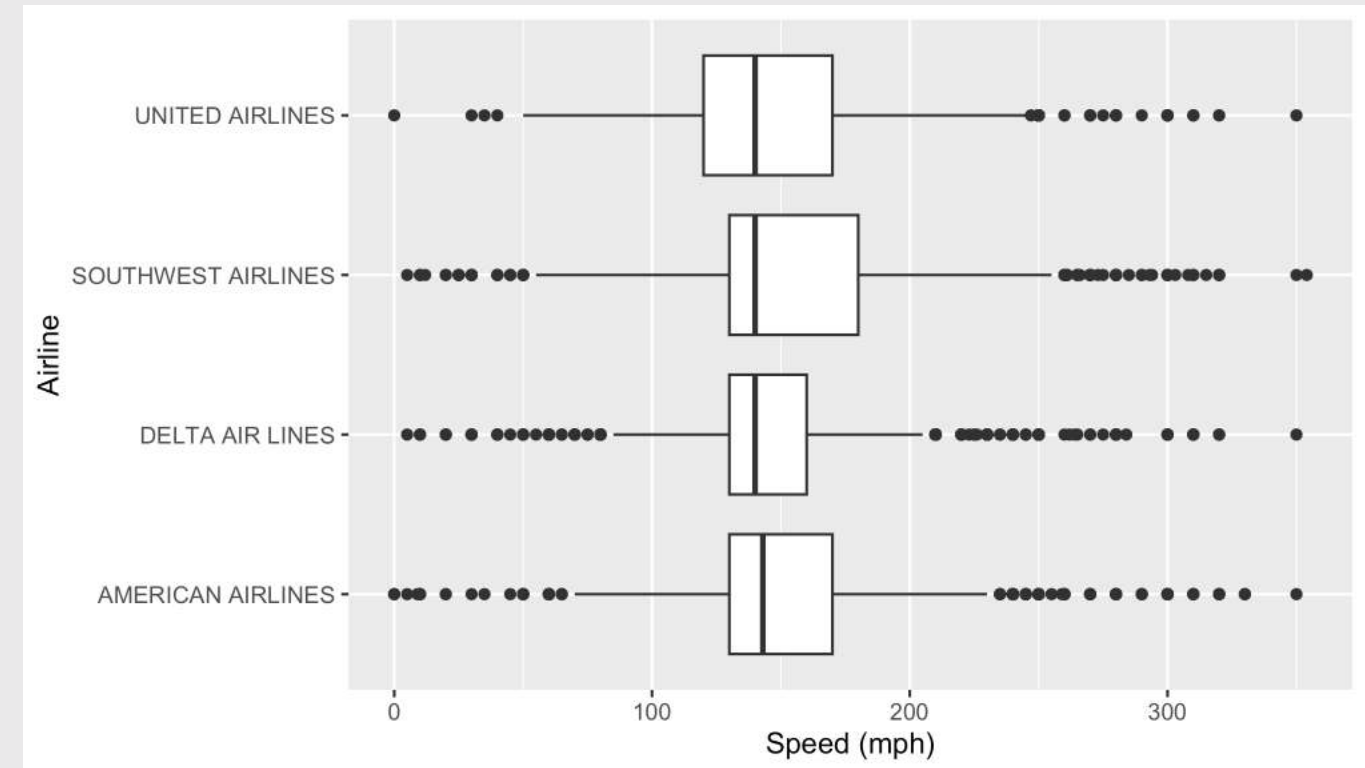
```
1 ggplot(wildlife_impacts) +  
2   geom_point(  
3     aes(  
4       x = speed,  
5       y = height  
6     ),  
7     size = 0.5  
8   ) +  
9   labs(  
10    x = 'Speed (mph)',  
11    y = 'Height (f)'  
12  )
```



One Continuous, One Categorical

Visualize with **boxplot**

```
1 ggplot(wildlife_impacts) +  
2   geom_boxplot(  
3     aes(  
4       x = speed,  
5       y = operator  
6     )  
7   ) +  
8   labs(  
9     x = 'Speed (mph)',  
10    y = 'Airline'  
11  )
```



Practice doing EDA

15:00

1. Read in the `candy_rankings.csv` data sets
2. Preview the data, note the data types and what each variable is.
3. Visualize (at least) three *relationships* between two variables (guided by a question) using an appropriate chart:
 - Bar chart
 - Scatterplot
 - Boxplot

Reminders:

- You have **4** days until your [Mini Project 1](#) is due (Oct. 4).
- Look at [HW 5](#), due next week